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Tissue visualization and manipulation system

Abstract

A tissue imaging assembly includes a steerable sheath and a steerable catheter deployable within the sheath. The tissue imaging assembly also includes an imaging hood at a distal end of the steerable catheter. The imaging hood is independently translatable with respect to the steerable sheath and the steerable catheter. The steerable catheter includes an imaging element and a lumen for delivering an elongate instrument out a distal region of the steerable catheter.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
623002	12/1898	Fitzgerald	N/A	N/A
623022	12/1898	Johnson	N/A	N/A
2305462	12/1941	Wolf	N/A	N/A
2453862	12/1947	Salisbury	N/A	N/A
3559651	12/1970	Moss	N/A	N/A
3831587	12/1973	Boyd	N/A	N/A
3874388	12/1974	King et al.	N/A	N/A
3903877	12/1974	Terada	N/A	N/A
4175545	12/1978	Termanini	N/A	N/A
4198981	12/1979	Sinnreich	N/A	N/A
4326529	12/1981	Doss et al.	N/A	N/A
4403612	12/1982	Fogarty	N/A	N/A
4445892	12/1983	Hussein et al.	N/A	N/A
4470407	12/1983	Hussein	N/A	N/A
4517976	12/1984	Murakoshi et al.	N/A	N/A
4569335	12/1985	Tsuno	N/A	N/A
4576146	12/1985	Kawazoe et al.	N/A	N/A
4615333	12/1985	Taguchi	N/A	N/A
4619247	12/1985	Inoue et al.	N/A	N/A
4676258	12/1986	Inokuchi et al.	N/A	N/A
4681093	12/1986	Ono et al.	N/A	N/A
4696668	12/1986	Wilcox	N/A	N/A
4709698	12/1986	Johnston et al.	N/A	N/A
4710192	12/1986	Liotta et al.	N/A	N/A

4727418	12/1987	Kato et al.	N/A	N/A
4772260	12/1987	Heyden	N/A	N/A
4784133	12/1987	Mackin	N/A	N/A
4838246	12/1988	Hahn et al.	N/A	N/A
4848323	12/1988	Marijnissen et al.	N/A	N/A
4880015	12/1988	Nierman	N/A	N/A
4911148	12/1989	Sosnowski et al.	N/A	N/A
4914521	12/1989	Adair	N/A	N/A
4943290	12/1989	Rexroth et al.	N/A	N/A
4950285	12/1989	Wilk	N/A	N/A
4957484	12/1989	Murtfeldt	N/A	N/A
4960411	12/1989	Buchbinder	N/A	N/A
4961738	12/1989	Mackin	N/A	N/A
4976710	12/1989	Mackin	N/A	N/A
4991578	12/1990	Cohen	N/A	N/A
4994069	12/1990	Ritchart et al.	N/A	N/A
4998916	12/1990	Hammerslag et al.	N/A	N/A
4998972	12/1990	Chin et al.	N/A	N/A
5025778	12/1990	Silverstein et al.	N/A	N/A
5047028	12/1990	Qian	N/A	N/A
5057106	12/1990	Kasevich et al.	N/A	N/A
5090959	12/1991	Samson et al.	N/A	N/A
5123428	12/1991	Schwarz	N/A	N/A
RE34002	12/1991	Adair	N/A	N/A
5156141	12/1991	Krebs et al.	N/A	N/A
5171259	12/1991	Inoue	N/A	N/A
5197457	12/1992	Adair	N/A	N/A
5270810	12/1992	Nishimura	348/241	H04N 5/335
5281238	12/1993	Chin et al.	N/A	N/A
5282827	12/1993	Kensey et al.	N/A	N/A
5305121	12/1993	Moll	N/A	N/A
5306234	12/1993	Johnson	N/A	N/A
5313934	12/1993	Wiita et al.	N/A	N/A
5313943	12/1993	Houser et al.	N/A	N/A
5330496	12/1993	Alferness	N/A	N/A
5334159	12/1993	Turkel	N/A	N/A
5334193	12/1993	Nardella	N/A	N/A
5336252	12/1993	Cohen	N/A	N/A
5339800	12/1993	Wiita et al.	N/A	N/A
5345927	12/1993	Bonutti	N/A	N/A
5348554	12/1993	Imran et al.	N/A	N/A
5353792	12/1993	Luebbers et al.	N/A	N/A
5370647	12/1993	Graber et al.	N/A	N/A
5373840	12/1993	Knighton	N/A	N/A
5375612	12/1993	Cottenceau et al.	N/A	N/A
5385146	12/1994	Goldreyer	N/A	N/A
5385148	12/1994	Lesh et al.	N/A	N/A
5391182	12/1994	Chin	N/A	N/A
5391199	12/1994	Ben-Haim	N/A	N/A

5403311	12/1994	Abele et al.	N/A	N/A
5403326	12/1994	Harrison et al.	N/A	N/A
5405360	12/1994	Tovey	N/A	N/A
5405376	12/1994	Mulier et al.	N/A	N/A
5409483	12/1994	Campbell et al.	N/A	N/A
5411016	12/1994	Kume et al.	N/A	N/A
5413104	12/1994	Buijs et al.	N/A	N/A
5421338	12/1994	Crowley et al.	N/A	N/A
5431649	12/1994	Mulier et al.	N/A	N/A
5453785	12/1994	Lenhardt et al.	N/A	N/A
5458612	12/1994	Chin	N/A	N/A
5462521	12/1994	Brucker et al.	N/A	N/A
5471515	12/1994	Fossum et al.	N/A	N/A
5498230	12/1995	Adair	N/A	N/A
5505730	12/1995	Edwards	N/A	N/A
5515853	12/1995	Smith et al.	N/A	N/A
5527338	12/1995	Purdy	N/A	N/A
5549603	12/1995	Feiring	N/A	N/A
5558619	12/1995	Kami et al.	N/A	N/A
5571088	12/1995	Lennox et al.	N/A	N/A
5575756	12/1995	Karasawa et al.	N/A	N/A
5575810	12/1995	Swanson et al.	N/A	N/A
5584872	12/1995	Lafontaine et al.	N/A	N/A
5591119	12/1996	Adair	N/A	N/A
5593405	12/1996	Osypka	N/A	N/A
5593422	12/1996	Muijs Van De Moer et al.	N/A	N/A
5593424	12/1996	Northrup, III	N/A	N/A
5643282	12/1996	Kieturakis	N/A	N/A
5653677	12/1996	Okada et al.	N/A	N/A
5662671	12/1996	Barbut et al.	N/A	N/A
5665062	12/1996	Houser	N/A	N/A
5672153	12/1996	Lax et al.	N/A	N/A
5676693	12/1996	Lafontaine	N/A	N/A
5681308	12/1996	Edwards et al.	N/A	N/A
5695448	12/1996	Kimura et al.	N/A	N/A
5697281	12/1996	Eggers et al.	N/A	N/A
5697882	12/1996	Eggers et al.	N/A	N/A
5709224	12/1997	Behl et al.	N/A	N/A
5713907	12/1997	Hogendijk et al.	N/A	N/A
5713946	12/1997	Ben-Haim	N/A	N/A
5716321	12/1997	Kerin et al.	N/A	N/A
5716325	12/1997	Bonutti	N/A	N/A
5722403	12/1997	McGee et al.	N/A	N/A
5725523	12/1997	Mueller	N/A	N/A
5743851	12/1997	Moll et al.	N/A	N/A
5746747	12/1997	McKeating	N/A	N/A
5749846	12/1997	Edwards et al.	N/A	N/A
5749889	12/1997	Bacich et al.	N/A	N/A
5749890	12/1997	Shaknovich	N/A	N/A

5754313	12/1997	Pelchy et al.	N/A	N/A
5766137	12/1997	Omata	N/A	N/A
5769846	12/1997	Edwards et al.	N/A	N/A
5792045	12/1997	Adair	N/A	N/A
5797903	12/1997	Swanson et al.	N/A	N/A
5823947	12/1997	Yoon	600/207	A61M 25/1011
5827175	12/1997	Tanaka et al.	N/A	N/A
5827268	12/1997	Laufer	N/A	N/A
5829447	12/1997	Stevens et al.	N/A	N/A
5842973	12/1997	Bullard	N/A	N/A
5843118	12/1997	Sepetka et al.	N/A	N/A
5846221	12/1997	Snoke et al.	N/A	N/A
5846239	12/1997	Swanson et al.	N/A	N/A
5848969	12/1997	Panescu et al.	N/A	N/A
5860974	12/1998	Abele	N/A	N/A
5860991	12/1998	Klein et al.	N/A	N/A
5865791	12/1998	Whayne et al.	N/A	N/A
5873815	12/1998	Kerin et al.	N/A	N/A
5879366	12/1998	Shaw et al.	N/A	N/A
5895417	12/1998	Pomeranz et al.	N/A	N/A
5897487	12/1998	Ouchi	N/A	N/A
5897553	12/1998	Mulier et al.	N/A	N/A
5902328	12/1998	LaFontaine et al.	N/A	N/A
5904651	12/1998	Swanson et al.	N/A	N/A
5908445	12/1998	Whayne et al.	N/A	N/A
5916147	12/1998	Boury	N/A	N/A
5925038	12/1998	Panescu et al.	N/A	N/A
5928250	12/1998	Koike et al.	N/A	N/A
5929901	12/1998	Adair et al.	N/A	N/A
5937614	12/1998	Watkins et al.	N/A	N/A
5941845	12/1998	Tu et al.	N/A	N/A
5944690	12/1998	Falwell et al.	N/A	N/A
5964755	12/1998	Edwards	N/A	N/A
5968053	12/1998	Revelas	N/A	N/A
5971983	12/1998	Lesh	N/A	N/A
5980484	12/1998	Ressemann et al.	N/A	N/A
5985307	12/1998	Hanson et al.	N/A	N/A
5986693	12/1998	Adair et al.	N/A	N/A
5997571	12/1998	Farr et al.	N/A	N/A
6004269	12/1998	Crowley et al.	N/A	N/A
6007521	12/1998	Bidwell et al.	N/A	N/A
6012457	12/1999	Lesh	N/A	N/A
6013024	12/1999	Mitsuda et al.	N/A	N/A
6024740	12/1999	Lesh et al.	N/A	N/A
6027501	12/1999	Goble et al.	N/A	N/A
6036685	12/1999	Mueller	N/A	N/A
6043839	12/1999	Adair et al.	N/A	N/A
6047218	12/1999	Whayne et al.	N/A	N/A
6063077	12/1999	Schaer	N/A	N/A

6063081	12/1999	Mulier et al.	N/A	N/A
6068653	12/1999	LaFontaine	N/A	N/A
6071279	12/1999	Whayne et al.	N/A	N/A
6071302	12/1999	Sinofsky et al.	N/A	N/A
6081740	12/1999	Gombrich et al.	N/A	N/A
6086528	12/1999	Adair	N/A	N/A
6086534	12/1999	Kesten	N/A	N/A
6099498	12/1999	Addis	N/A	N/A
6099514	12/1999	Sharkey et al.	N/A	N/A
6102905	12/1999	Baxter et al.	N/A	N/A
6112123	12/1999	Kelleher et al.	N/A	N/A
6115626	12/1999	Whayne et al.	N/A	N/A
6123703	12/1999	Tu et al.	N/A	N/A
6123718	12/1999	Tu et al.	N/A	N/A
6129724	12/1999	Fleischman et al.	N/A	N/A
6139508	12/1999	Simpson et al.	N/A	N/A
6142993	12/1999	Whayne et al.	N/A	N/A
6152144	12/1999	Lesh et al.	N/A	N/A
6156350	12/1999	Constantz	N/A	N/A
6159203	12/1999	Sinofsky	N/A	N/A
6161543	12/1999	Cox et al.	N/A	N/A
6164283	12/1999	Lesh	N/A	N/A
6167297	12/1999	Benaron	N/A	N/A
6168591	12/2000	Sinofsky	N/A	N/A
6168594	12/2000	LaFontaine et al.	N/A	N/A
6174307	12/2000	Daniel et al.	N/A	N/A
6178346	12/2000	Amundson et al.	N/A	N/A
6190381	12/2000	Olsen et al.	N/A	N/A
6211904	12/2000	Adair et al.	N/A	N/A
6224553	12/2000	Nevo	N/A	N/A
6231561	12/2000	Frazier et al.	N/A	N/A
6234995	12/2000	Peacock, III	N/A	N/A
6235044	12/2000	Root et al.	N/A	N/A
6237605	12/2000	Vaska et al.	N/A	N/A
6238393	12/2000	Mulier et al.	N/A	N/A
6240312	12/2000	Alfano et al.	N/A	N/A
6254598	12/2000	Edwards et al.	N/A	N/A
6258083	12/2000	Daniel et al.	N/A	N/A
6261226	12/2000	McKenna et al.	N/A	N/A
6263224	12/2000	West	N/A	N/A
6266551	12/2000	Osadchy et al.	N/A	N/A
6270492	12/2000	Sinofsky	N/A	N/A
6275255	12/2000	Adair et al.	N/A	N/A
6280450	12/2000	McGuckin, Jr.	N/A	N/A
6290689	12/2000	Delaney et al.	N/A	N/A
6306081	12/2000	Ishikawa et al.	N/A	N/A
6310642	12/2000	Adair et al.	N/A	N/A
6311692	12/2000	Vaska et al.	N/A	N/A
6314962	12/2000	Vaska et al.	N/A	N/A
6314963	12/2000	Vaska et al.	N/A	N/A

6315777	12/2000	Comben	N/A	N/A
6315778	12/2000	Gambale et al.	N/A	N/A
6322536	12/2000	Rosengart et al.	N/A	N/A
6325797	12/2000	Stewart et al.	N/A	N/A
6328727	12/2000	Frazier et al.	N/A	N/A
6358247	12/2001	Altman et al.	N/A	N/A
6358248	12/2001	Mulier et al.	N/A	N/A
6375654	12/2001	McIntyre	N/A	N/A
6379345	12/2001	Constantz	N/A	N/A
6383195	12/2001	Richard	N/A	N/A
6385476	12/2001	Osadchy et al.	N/A	N/A
6387043	12/2001	Yoon	N/A	N/A
6387071	12/2001	Constantz	N/A	N/A
6394096	12/2001	Constantz	N/A	N/A
6396873	12/2001	Goldstein et al.	N/A	N/A
6398780	12/2001	Farley et al.	N/A	N/A
6401719	12/2001	Farley et al.	N/A	N/A
6409722	12/2001	Hoey et al.	N/A	N/A
6416511	12/2001	Lesh et al.	N/A	N/A
6419669	12/2001	Frazier et al.	N/A	N/A
6423051	12/2001	Kaplan et al.	N/A	N/A
6423055	12/2001	Farr et al.	N/A	N/A
6423058	12/2001	Edwards et al.	N/A	N/A
6428536	12/2001	Panescu et al.	N/A	N/A
6436118	12/2001	Kayan	N/A	N/A
6440061	12/2001	Wenner et al.	N/A	N/A
6440119	12/2001	Nakada et al.	N/A	N/A
6458151	12/2001	Saltiel	N/A	N/A
6461327	12/2001	Addis et al.	N/A	N/A
6464697	12/2001	Edwards et al.	N/A	N/A
6474340	12/2001	Vaska et al.	N/A	N/A
6475223	12/2001	Werp et al.	N/A	N/A
6478769	12/2001	Parker	N/A	N/A
6482162	12/2001	Moore	N/A	N/A
6484727	12/2001	Vaska et al.	N/A	N/A
6485489	12/2001	Teirstein et al.	N/A	N/A
6488671	12/2001	Constantz et al.	N/A	N/A
6494902	12/2001	Hoey et al.	N/A	N/A
6497651	12/2001	Kan et al.	N/A	N/A
6497705	12/2001	Comben	N/A	N/A
6500174	12/2001	Maguire et al.	N/A	N/A
6502576	12/2002	Lesh	N/A	N/A
6514249	12/2002	Maguire et al.	N/A	N/A
6517533	12/2002	Swaminathan	N/A	N/A
6527979	12/2002	Constantz et al.	N/A	N/A
6532380	12/2002	Close et al.	N/A	N/A
6533767	12/2002	Johansson et al.	N/A	N/A
6537272	12/2002	Christopherson et al.	N/A	N/A
6538375	12/2002	Duggal et al.	N/A	N/A

6540733	12/2002	Constantz et al.	N/A	N/A
6540744	12/2002	Hassett et al.	N/A	N/A
6544195	12/2002	Wilson et al.	N/A	N/A
6547780	12/2002	Sinofsky	N/A	N/A
6549800	12/2002	Atalar et al.	N/A	N/A
6558375	12/2002	Sinofsky et al.	N/A	N/A
6558382	12/2002	Jahns et al.	N/A	N/A
6562020	12/2002	Constantz et al.	N/A	N/A
6572609	12/2002	Farr et al.	N/A	N/A
6579285	12/2002	Sinofsky	N/A	N/A
6585732	12/2002	Mulier et al.	N/A	N/A
6587709	12/2002	Solf et al.	N/A	N/A
6593884	12/2002	Gilboa et al.	N/A	N/A
6605055	12/2002	Sinofsky et al.	N/A	N/A
6613062	12/2002	Leckrone et al.	N/A	N/A
6622732	12/2002	Constantz	N/A	N/A
6626855	12/2002	Weng et al.	N/A	N/A
6626899	12/2002	Houser et al.	N/A	N/A
6626900	12/2002	Sinofsky et al.	N/A	N/A
6632171	12/2002	Iddan et al.	N/A	N/A
6635070	12/2002	Leeflang et al.	N/A	N/A
6645202	12/2002	Pless et al.	N/A	N/A
6650923	12/2002	Lesh et al.	N/A	N/A
6658279	12/2002	Swanson et al.	N/A	N/A
6659940	12/2002	Adler	N/A	N/A
6663821	12/2002	Seward	N/A	N/A
6673090	12/2003	Root et al.	N/A	N/A
6676656	12/2003	Sinofsky	N/A	N/A
6679836	12/2003	Couvillon, Jr. et al.	N/A	N/A
6682526	12/2003	Jones et al.	N/A	N/A
6689051	12/2003	Nakada et al.	N/A	N/A
6689128	12/2003	Sliwa et al.	N/A	N/A
6692430	12/2003	Adler	N/A	N/A
6701581	12/2003	Senovich et al.	N/A	N/A
6701931	12/2003	Sliwa et al.	N/A	N/A
6702780	12/2003	Gilboa et al.	N/A	N/A
6704043	12/2003	Goldstein et al.	N/A	N/A
6706039	12/2003	Mulier et al.	N/A	N/A
6712798	12/2003	Constantz	N/A	N/A
6719747	12/2003	Constantz et al.	N/A	N/A
6719755	12/2003	Sliwa et al.	N/A	N/A
6730063	12/2003	Delaney et al.	N/A	N/A
6736810	12/2003	Hoey et al.	N/A	N/A
6749617	12/2003	Palasis et al.	N/A	N/A
6751492	12/2003	Ben-Haim	N/A	N/A
6755790	12/2003	Stewart et al.	N/A	N/A
6755811	12/2003	Constantz	N/A	N/A
6764487	12/2003	Mulier et al.	N/A	N/A
6770070	12/2003	Balbierz	N/A	N/A
6771996	12/2003	Bowe et al.	N/A	N/A

6773402	12/2003	Govari et al.	N/A	N/A
6780151	12/2003	Grabover et al.	N/A	N/A
6805128	12/2003	Pless et al.	N/A	N/A
6805129	12/2003	Pless et al.	N/A	N/A
6811562	12/2003	Pless	N/A	N/A
6833814	12/2003	Gilboa et al.	N/A	N/A
6840923	12/2004	Lapcevic	N/A	N/A
6840936	12/2004	Sliwa et al.	N/A	N/A
6849073	12/2004	Hoey et al.	N/A	N/A
6858005	12/2004	Ohline et al.	N/A	N/A
6858026	12/2004	Sliwa et al.	N/A	N/A
6858905	12/2004	Hsu et al.	N/A	N/A
6863668	12/2004	Gillespie et al.	N/A	N/A
6866651	12/2004	Constantz	N/A	N/A
6887237	12/2004	McGaffigan	N/A	N/A
6892091	12/2004	Ben-Haim et al.	N/A	N/A
6896690	12/2004	Lambrecht et al.	N/A	N/A
6899672	12/2004	Chin et al.	N/A	N/A
6915154	12/2004	Docherty et al.	N/A	N/A
6916284	12/2004	Moriyama	N/A	N/A
6916286	12/2004	Kazakevich	N/A	N/A
6923805	12/2004	LaFontaine et al.	N/A	N/A
6929010	12/2004	Vaska et al.	N/A	N/A
6932809	12/2004	Sinofsky	N/A	N/A
6939348	12/2004	Malecki et al.	N/A	N/A
6942657	12/2004	Sinofsky et al.	N/A	N/A
6949095	12/2004	Vaska et al.	N/A	N/A
6953457	12/2004	Farr et al.	N/A	N/A
6955173	12/2004	Lesh	N/A	N/A
6958069	12/2004	Shipp, I et al.	N/A	N/A
6962589	12/2004	Mulier et al.	N/A	N/A
6971394	12/2004	Sliwa et al.	N/A	N/A
6974464	12/2004	Quijano et al.	N/A	N/A
6979290	12/2004	Mourlas et al.	N/A	N/A
6982740	12/2005	Adair et al.	N/A	N/A
6984232	12/2005	Vanney et al.	N/A	N/A
6994094	12/2005	Schwartz	N/A	N/A
7001329	12/2005	Kobayashi et al.	N/A	N/A
7019610	12/2005	Creighton et al.	N/A	N/A
7025746	12/2005	Tal	N/A	N/A
7030904	12/2005	Adair et al.	N/A	N/A
7041098	12/2005	Farley et al.	N/A	N/A
7042487	12/2005	Nakashima	N/A	N/A
7044135	12/2005	Lesh	N/A	N/A
7052493	12/2005	Vaska et al.	N/A	N/A
7090683	12/2005	Brock et al.	N/A	N/A
7118566	12/2005	Jahns	N/A	N/A
7156845	12/2006	Mulier et al.	N/A	N/A
7163534	12/2006	Brucker et al.	N/A	N/A
7166537	12/2006	Jacobsen et al.	N/A	N/A

7169144	12/2006	Hoey et al.	N/A	N/A
7179224	12/2006	Willis	N/A	N/A
7186214	12/2006	Ness	N/A	N/A
7207984	12/2006	Farr et al.	N/A	N/A
7217268	12/2006	Eggers et al.	N/A	N/A
7242832	12/2006	Carlin et al.	N/A	N/A
7247155	12/2006	Hoey et al.	N/A	N/A
7261711	12/2006	Mulier et al.	N/A	N/A
7263397	12/2006	Hauck et al.	N/A	N/A
7276061	12/2006	Schaer et al.	N/A	N/A
7309328	12/2006	Kaplan et al.	N/A	N/A
7322934	12/2007	Miyake et al.	N/A	N/A
7323001	12/2007	Clubb et al.	N/A	N/A
7416552	12/2007	Paul et al.	N/A	N/A
7435248	12/2007	Taimisto et al.	N/A	N/A
7527625	12/2008	Knight et al.	N/A	N/A
7534204	12/2008	Starksen et al.	N/A	N/A
7534294	12/2008	Gaynor et al.	N/A	N/A
7569052	12/2008	Phan et al.	N/A	N/A
7569952	12/2008	Bono et al.	N/A	N/A
7736347	12/2009	Kaplan et al.	N/A	N/A
7758499	12/2009	Adler	N/A	N/A
7860555	12/2009	Saadat	N/A	N/A
7860556	12/2009	Saadat	N/A	N/A
7918787	12/2010	Saadat	N/A	N/A
7919610	12/2010	Serebriiskii et al.	N/A	N/A
7930016	12/2010	Saadat	N/A	N/A
8050746	12/2010	Saadat et al.	N/A	N/A
8078266	12/2010	Saadat et al.	N/A	N/A
8131350	12/2011	Saadat et al.	N/A	N/A
8137333	12/2011	Saadat et al.	N/A	N/A
8221310	12/2011	Saadat et al.	N/A	N/A
8235985	12/2011	Saadat et al.	N/A	N/A
8333012	12/2011	Rothe et al.	N/A	N/A
8417321	12/2012	Saadat et al.	N/A	N/A
8419613	12/2012	Saadat et al.	N/A	N/A
8475361	12/2012	Barlow et al.	N/A	N/A
8657805	12/2013	Peh et al.	N/A	N/A
8758229	12/2013	Saadat et al.	N/A	N/A
8814845	12/2013	Saadat et al.	N/A	N/A
8906007	12/2013	Bonn et al.	N/A	N/A
8934962	12/2014	Saadat et al.	N/A	N/A
9055906	12/2014	Saadat et al.	N/A	N/A
9155587	12/2014	Willis et al.	N/A	N/A
9192287	12/2014	Saadat et al.	N/A	N/A
9226648	12/2015	Saadat et al.	N/A	N/A
9332893	12/2015	Saadat et al.	N/A	N/A
9510732	12/2015	Miller et al.	N/A	N/A
9526401	12/2015	Saadat et al.	N/A	N/A
10004388	12/2017	Saadat et al.	N/A	N/A

10064540	12/2017	Saadat et al.	N/A	N/A
10070772	12/2017	Peh et al.	N/A	N/A
10092172	12/2017	Peh et al.	N/A	N/A
10278588	12/2018	Saadat et al.	N/A	N/A
10368729	12/2018	Miller et al.	N/A	N/A
10390685	12/2018	Saadat et al.	N/A	N/A
10463237	12/2018	Saadat et al.	N/A	N/A
10470643	12/2018	Saadat et al.	N/A	N/A
10555788	12/2019	Panescu et al.	N/A	N/A
10772492	12/2019	Miller et al.	N/A	N/A
11304771	12/2021	Panescu et al.	N/A	N/A
11337594	12/2021	Saadat et al.	N/A	N/A
11406250	12/2021	Saadat et al.	N/A	N/A
11478152	12/2021	Saadat et al.	N/A	N/A
2001/0020126	12/2000	Swanson et al.	N/A	N/A
2001/0039416	12/2000	Moorman et al.	N/A	N/A
2001/0047136	12/2000	Domanik et al.	N/A	N/A
2001/0047184	12/2000	Connors	N/A	N/A
2002/0004644	12/2001	Koblish	N/A	N/A
2002/0026145	12/2001	Bagaoisan et al.	N/A	N/A
2002/0035311	12/2001	Ouchi	N/A	N/A
2002/0054852	12/2001	Cate	N/A	N/A
2002/0065455	12/2001	Ben-Haim Shlomo et al.	N/A	N/A
2002/0072710	12/2001	Stewart	604/164.02	A61M 25/0662
2002/0077594	12/2001	Chien et al.	N/A	N/A
2002/0077642	12/2001	Patel et al.	N/A	N/A
2002/0087169	12/2001	Brock et al.	N/A	N/A
2002/0091304	12/2001	Ogura et al.	N/A	N/A
2002/0138088	12/2001	Nash et al.	N/A	N/A
2002/0161377	12/2001	Rabkin et al.	N/A	N/A
2002/0165598	12/2001	Wahr et al.	N/A	N/A
2002/0169377	12/2001	Khairkhahan et al.	N/A	N/A
2002/0173815	12/2001	Hogendijk et al.	N/A	N/A
2003/0009085	12/2002	Arai et al.	N/A	N/A
2003/0014010	12/2002	Carpenter et al.	N/A	N/A
2003/0018358	12/2002	Saadat	N/A	N/A
2003/0035156	12/2002	Cooper	N/A	N/A
2003/0036698	12/2002	Kohler et al.	N/A	N/A
2003/0065267	12/2002	Smith	N/A	N/A
2003/0069593	12/2002	Tremulis et al.	N/A	N/A
2003/0120142	12/2002	Dubuc et al.	N/A	N/A
2003/0130572	12/2002	Phan et al.	N/A	N/A
2003/0144657	12/2002	Bowe et al.	N/A	N/A
2003/0171741	12/2002	Ziebol et al.	N/A	N/A
2003/0181939	12/2002	Bonutti	N/A	N/A
2003/0208222	12/2002	Zadno-Azizi Gholam-Reza	N/A	N/A
2003/0212394	12/2002	Pearson et al.	N/A	N/A

2003/0220574	12/2002	Markus et al.	N/A	N/A
2003/0222325	12/2002	Jacobsen et al.	N/A	N/A
2003/0236493	12/2002	Mauch	N/A	N/A
2004/0044350	12/2003	Martin et al.	N/A	N/A
2004/0049211	12/2003	Tremulis et al.	N/A	N/A
2004/0054335	12/2003	Lesh et al.	N/A	N/A
2004/0054389	12/2003	Osypka	N/A	N/A
2004/0082833	12/2003	Adler et al.	N/A	N/A
2004/0097788	12/2003	Mourlas	600/116	A61B 1/05
2004/0097792	12/2003	Moll et al.	N/A	N/A
2004/0097805	12/2003	Verard et al.	N/A	N/A
2004/0098031	12/2003	Van Der Burg et al.	N/A	N/A
2004/0117032	12/2003	Roth	N/A	N/A
2004/0133113	12/2003	Krishnan	N/A	N/A
2004/0138529	12/2003	Wiltshire et al.	N/A	N/A
2004/0138707	12/2003	Greenhalgh	N/A	N/A
2004/0147806	12/2003	Adler	N/A	N/A
2004/0147911	12/2003	Sinofsky	N/A	N/A
2004/0147912	12/2003	Sinofsky	N/A	N/A
2004/0147913	12/2003	Sinofsky	N/A	N/A
2004/0158143	12/2003	Flaherty et al.	N/A	N/A
2004/0158289	12/2003	Girouard et al.	N/A	N/A
2004/0165766	12/2003	Goto	N/A	N/A
2004/0167503	12/2003	Sinofsky	N/A	N/A
2004/0181237	12/2003	Forde et al.	N/A	N/A
2004/0199052	12/2003	Banik et al.	N/A	N/A
2004/0210111	12/2003	Okada	N/A	N/A
2004/0210239	12/2003	Nash et al.	N/A	N/A
2004/0210278	12/2003	Boll et al.	N/A	N/A
2004/0215180	12/2003	Starkebaum et al.	N/A	N/A
2004/0220471	12/2003	Schwartz	N/A	N/A
2004/0230131	12/2003	Kassab et al.	N/A	N/A
2004/0248837	12/2003	Raz et al.	N/A	N/A
2004/0249367	12/2003	Saadat et al.	N/A	N/A
2004/0254523	12/2003	Fitzgerald et al.	N/A	N/A
2004/0260182	12/2003	Zuluaga et al.	N/A	N/A
2004/0267084	12/2003	Navia et al.	N/A	N/A
2005/0004597	12/2004	McGuckin, Jr. et al.	N/A	N/A
2005/0014995	12/2004	Amundson et al.	N/A	N/A
2005/0015048	12/2004	Chiu et al.	N/A	N/A
2005/0020914	12/2004	Amundson et al.	N/A	N/A
2005/0027163	12/2004	Chin et al.	N/A	N/A
2005/0038419	12/2004	Arnold et al.	N/A	N/A
2005/0059862	12/2004	Phan	N/A	N/A
2005/0059954	12/2004	Constantz	N/A	N/A
2005/0059965	12/2004	Eberl et al.	N/A	N/A
2005/0059984	12/2004	Chanduszko et al.	N/A	N/A
2005/0065504	12/2004	Melsky et al.	N/A	N/A
2005/0080336	12/2004	Byrd et al.	N/A	N/A

2005/0090818	12/2004	Pike, Jr. et al.	N/A	N/A
2005/0096502	12/2004	Khalili	N/A	N/A
2005/0096643	12/2004	Brucker et al.	N/A	N/A
2005/0101984	12/2004	Chanduszko et al.	N/A	N/A
2005/0107736	12/2004	Landman et al.	N/A	N/A
2005/0124969	12/2004	Fitzgerald et al.	N/A	N/A
2005/0131401	12/2004	Malecki et al.	N/A	N/A
2005/0154252	12/2004	Sharkey et al.	N/A	N/A
2005/0159702	12/2004	Sekiguchi et al.	N/A	N/A
2005/0165272	12/2004	Okada et al.	N/A	N/A
2005/0165279	12/2004	Adler et al.	N/A	N/A
2005/0165391	12/2004	Maguire et al.	N/A	N/A
2005/0165466	12/2004	Morris et al.	N/A	N/A
2005/0197530	12/2004	Wallace et al.	N/A	N/A
2005/0197623	12/2004	Leeflang et al.	N/A	N/A
2005/0215895	12/2004	Popp et al.	N/A	N/A
2005/0222554	12/2004	Wallace et al.	N/A	N/A
2005/0222557	12/2004	Baxter et al.	N/A	N/A
2005/0222558	12/2004	Baxter et al.	N/A	N/A
2005/0228452	12/2004	Mourlas et al.	N/A	N/A
2005/0234436	12/2004	Baxter et al.	N/A	N/A
2005/0234437	12/2004	Baxter et al.	N/A	N/A
2005/0267328	12/2004	Blumzvig et al.	N/A	N/A
2005/0288632	12/2004	Willard	N/A	N/A
2006/0009715	12/2005	Khairkhahan et al.	N/A	N/A
2006/0009737	12/2005	Whiting et al.	N/A	N/A
2006/0015096	12/2005	Hauck et al.	N/A	N/A
2006/0022234	12/2005	Adair et al.	N/A	N/A
2006/0025651	12/2005	Adler et al.	N/A	N/A
2006/0025787	12/2005	Morales et al.	N/A	N/A
2006/0058598	12/2005	Esposito	N/A	N/A
2006/0069303	12/2005	Couvillon, Jr.	N/A	N/A
2006/0069313	12/2005	Couvillon, Jr. et al.	N/A	N/A
2006/0074398	12/2005	Whiting et al.	N/A	N/A
2006/0084839	12/2005	Mourlas et al.	N/A	N/A
2006/0084945	12/2005	Moll et al.	N/A	N/A
2006/0089637	12/2005	Werneth et al.	N/A	N/A
2006/0111614	12/2005	Saadat et al.	N/A	N/A
2006/0111692	12/2005	Hlavka et al.	N/A	N/A
2006/0122587	12/2005	Sharareh	N/A	N/A
2006/0146172	12/2005	Jacobsen et al.	N/A	N/A
2006/0149129	12/2005	Watts et al.	N/A	N/A
2006/0149331	12/2005	Mann et al.	N/A	N/A
2006/0155242	12/2005	Constantz	N/A	N/A
2006/0161133	12/2005	Laird et al.	N/A	N/A
2006/0167439	12/2005	Kalser et al.	N/A	N/A
2006/0183992	12/2005	Kawashima	N/A	N/A
2006/0195060	12/2005	Navia et al.	N/A	N/A
2006/0217755	12/2005	Eversull et al.	N/A	N/A
2006/0224167	12/2005	Weisenburgh et al.	N/A	N/A

2006/0253113	12/2005	Arnold et al.	N/A	N/A
2006/0258909	12/2005	Saadat et al.	N/A	N/A
2006/0271032	12/2005	Chin et al.	N/A	N/A
2007/0005019	12/2006	Okishige	N/A	N/A
2007/0015964	12/2006	Eversull et al.	N/A	N/A
2007/0016130	12/2006	Leeflang et al.	N/A	N/A
2007/0043338	12/2006	Moll et al.	N/A	N/A
2007/0043413	12/2006	Eversull et al.	N/A	N/A
2007/0049923	12/2006	Jahns	N/A	N/A
2007/0055142	12/2006	Webler	N/A	N/A
2007/0078451	12/2006	Arnold et al.	N/A	N/A
2007/0083099	12/2006	Henderson et al.	N/A	N/A
2007/0083187	12/2006	Eversull et al.	N/A	N/A
2007/0083217	12/2006	Eversull et al.	N/A	N/A
2007/0093808	12/2006	Mulier et al.	N/A	N/A
2007/0100324	12/2006	Tempel et al.	N/A	N/A
2007/0106113	12/2006	Ravo	N/A	N/A
2007/0106146	12/2006	Altmann et al.	N/A	N/A
2007/0106214	12/2006	Gray et al.	N/A	N/A
2007/0106287	12/2006	O'Sullivan	N/A	N/A
2007/0135826	12/2006	Zaver et al.	N/A	N/A
2007/0167801	12/2006	Webler et al.	N/A	N/A
2007/0213584	12/2006	Kim et al.	N/A	N/A
2007/0239010	12/2006	Johnson	N/A	N/A
2007/0265609	12/2006	Thapliyal et al.	N/A	N/A
2007/0265610	12/2006	Thapliyal et al.	N/A	N/A
2007/0270639	12/2006	Long	N/A	N/A
2007/0270686	12/2006	Ritter et al.	N/A	N/A
2007/0282371	12/2006	Lee et al.	N/A	N/A
2007/0299456	12/2006	Teague	N/A	N/A
2008/0009747	12/2007	Saadat et al.	N/A	N/A
2008/0009859	12/2007	Auth et al.	N/A	N/A
2008/0015563	12/2007	Hoey et al.	N/A	N/A
2008/0015569	12/2007	Saadat et al.	N/A	N/A
2008/0027464	12/2007	Moll et al.	N/A	N/A
2008/0033241	12/2007	Peh et al.	N/A	N/A
2008/0057106	12/2007	Erickson et al.	N/A	N/A
2008/0058590	12/2007	Saadat et al.	N/A	N/A
2008/0058836	12/2007	Moll et al.	N/A	N/A
2008/0097476	12/2007	Peh et al.	N/A	N/A
2008/0183081	12/2007	Lys et al.	N/A	N/A
2008/0214889	12/2007	Saadat et al.	N/A	N/A
2008/0228032	12/2007	Starksen et al.	N/A	N/A
2008/0234834	12/2007	Meade et al.	N/A	N/A
2008/0275300	12/2007	Rothe et al.	N/A	N/A
2008/0287790	12/2007	Li	N/A	N/A
2008/0287805	12/2007	Li	N/A	N/A
2008/0319258	12/2007	Thompson	N/A	N/A
2009/0030276	12/2008	Saadat et al.	N/A	N/A
2009/0030412	12/2008	Willis et al.	N/A	N/A

2009/0048480	12/2008	Klenk et al.	N/A	N/A
2009/0054805	12/2008	Boyle, Jr.	N/A	N/A
2009/0062790	12/2008	Malchano et al.	N/A	N/A
2009/0062871	12/2008	Chin et al.	N/A	N/A
2009/0076476	12/2008	Barbagli et al.	N/A	N/A
2009/0076489	12/2008	Welches et al.	N/A	N/A
2009/0082623	12/2008	Rothe et al.	N/A	N/A
2009/0125022	12/2008	Saadat et al.	N/A	N/A
2009/0143640	12/2008	Saadat et al.	N/A	N/A
2009/0187074	12/2008	Saadat et al.	N/A	N/A
2009/0203962	12/2008	Miller et al.	N/A	N/A
2009/0227999	12/2008	Willis et al.	N/A	N/A
2009/0264727	12/2008	Markowitz et al.	N/A	N/A
2009/0267773	12/2008	Markowitz et al.	N/A	N/A
2009/0326572	12/2008	Peh et al.	N/A	N/A
2010/0004506	12/2009	Saadat	N/A	N/A
2010/0004633	12/2009	Rothe et al.	N/A	N/A
2010/0004661	12/2009	Verin et al.	N/A	N/A
2010/0130836	12/2009	Malchano et al.	N/A	N/A
2011/0021926	12/2010	Spencer et al.	N/A	N/A
2011/0060227	12/2010	Saadat	N/A	N/A
2011/0060298	12/2010	Saadat	N/A	N/A
2011/0144576	12/2010	Rothe et al.	N/A	N/A
2011/0196237	12/2010	Pelissier et al.	N/A	N/A
2012/0016221	12/2011	Saadat et al.	N/A	N/A
2012/0059366	12/2011	Drews et al.	N/A	N/A
2012/0095332	12/2011	Nitta et al.	N/A	N/A
2012/0150046	12/2011	Watson et al.	N/A	N/A
2013/0172745	12/2012	Choi	N/A	N/A
2014/0012074	12/2013	Vazales et al.	N/A	N/A
2015/0094577	12/2014	Saadat et al.	N/A	N/A
2015/0094582	12/2014	Tanaka et al.	N/A	N/A
2015/0190036	12/2014	Saadat	N/A	N/A
2015/0250382	12/2014	Saadat et al.	N/A	N/A
2015/0366440	12/2014	Rothe	600/104	A61B 1/05
2016/0038005	12/2015	Saadat et al.	N/A	N/A
2016/0095501	12/2015	Saadat et al.	N/A	N/A
2016/0227989	12/2015	Saadat et al.	N/A	N/A
2016/0361040	12/2015	Tanaka et al.	N/A	N/A
2017/0071460	12/2016	Miller et al.	N/A	N/A
2018/0000314	12/2017	Saadat et al.	N/A	N/A
2018/0228350	12/2017	Saadat et al.	N/A	N/A
2019/0008360	12/2018	Peh et al.	N/A	N/A
2019/0014975	12/2018	Saadat et al.	N/A	N/A
2019/0021577	12/2018	Peh et al.	N/A	N/A
2019/0046013	12/2018	Saadat et al.	N/A	N/A
2019/0307331	12/2018	Saadat et al.	N/A	N/A
2019/0343373	12/2018	Saadat et al.	N/A	N/A
2020/0000319	12/2019	Saadat	N/A	N/A
2020/0054200	12/2019	Saadat et al.	N/A	N/A

2020/0069166	12/2019	Miller et al.	N/A	N/A
2020/0305693	12/2019	Saadat et al.	N/A	N/A
2021/0007594	12/2020	Miller et al.	N/A	N/A
2022/0338712	12/2021	Saadat et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2853466	12/1978	DE	N/A
10028155	12/1999	DE	N/A
0283661	12/1987	EP	N/A
0301288	12/1988	EP	N/A
0842673	12/1997	EP	N/A
S5993413	12/1983	JP	N/A
S59181315	12/1983	JP	N/A
H01221133	12/1988	JP	N/A
H03284265	12/1990	JP	N/A
H05103746	12/1992	JP	N/A
H06507809	12/1993	JP	N/A
H0951897	12/1996	JP	N/A
H11299725	12/1998	JP	N/A
2001504363	12/2000	JP	N/A
2001258822	12/2000	JP	N/A
WO-9221292	12/1991	WO	N/A
WO-9407413	12/1993	WO	N/A
WO-9503843	12/1994	WO	N/A
WO-9740880	12/1996	WO	N/A
WO-9818388	12/1997	WO	N/A
WO-0024310	12/1999	WO	N/A
WO-0149356	12/2000	WO	N/A
WO-0172368	12/2000	WO	N/A
WO-0230310	12/2001	WO	N/A
WO-03037416	12/2002	WO	N/A
WO-03039350	12/2002	WO	N/A
WO-03053491	12/2002	WO	N/A
WO-03073942	12/2002	WO	N/A
WO-03101287	12/2002	WO	N/A
WO-2004043272	12/2003	WO	N/A
WO-2004080508	12/2003	WO	N/A
WO-2005070330	12/2004	WO	N/A
WO-2005077435	12/2004	WO	N/A
WO-2005081202	12/2004	WO	N/A
WO-2006017517	12/2005	WO	N/A
WO-2006024015	12/2005	WO	N/A
WO-2006083794	12/2005	WO	N/A
WO-2006091597	12/2005	WO	N/A
WO-2006126979	12/2005	WO	N/A
WO-2007067323	12/2006	WO	N/A
WO-2007079268	12/2006	WO	N/A
WO-2007133845	12/2006	WO	N/A

WO-2007134258	12/2006	WO	N/A
WO-2008015625	12/2007	WO	N/A
WO-2008021994	12/2007	WO	N/A
WO-2008021997	12/2007	WO	N/A
WO-2008021998	12/2007	WO	N/A
WO-2008024261	12/2007	WO	N/A
WO-2008079828	12/2007	WO	N/A
WO-2009112262	12/2008	WO	N/A

OTHER PUBLICATIONS

Avitall B., et al., "Right-Sided Driven Atrial Fibrillation in a Sterile Pericarditis Dog Model," Pacing and Clinical Electrophysiology, 1994, vol. 17, pp. 774. cited by applicant

Avitall, et al. "A Catheter System to Ablate Atrial Fibrillation in a Sterile Pericarditis Dog Model," Pacing and Clinical Electrophysiology, 1994, vol. 17, pp. 774. cited by applicant

Avitall, "Vagally Mediated Atrial Fibrillation in a Dog Model can be Ablated by Placing Linear Radiofrequency Lesions at the Junction of the Right Atrial Appendage and the Superior Vena Cava," Pacing and Clinical Electrophysiology, 1995, vol. 18, pp. 857. cited by applicant

Baker B.M., et al., "Nonpharmacologic Approaches to the Treatment of Atrial Fibrillation and Atrial Flutter," Journal of Cardiovascular Electrophysiology, 1995, vol. 6 (10 Pt 2), pp. 972-978. cited by applicant

Bhakta D., et al., "Principles of Electroanatomic Mapping," Indian Pacing and Electrophysiology Journal, 2008, vol. 8 (1), pp. 32-50. cited by applicant

Bidoggia H., et al., "Transseptal Left Heart Catheterization: Usefulness of the Intracavitary Electrocardiogram in the Localization of the Fossa Ovalis," Cathet Cardiovasc Diagn, 1991, vol. 24 (3), pp. 221-225, PMID: 1764747 [online], [retrieved Feb. 15, 2010]. Retrieved from the Internet:< URL: <http://www.ncbi.nlm.nih.gov/sites/entrez>>. cited by applicant

Bredikis J.J., et al., "Surgery of Tachyarrhythmia: Intracardiac Closed Heart Cryoablation," Pacing and Clinical Electrophysiology, 1990, vol. 13 (Part 2), pp. 1980-1984. cited by applicant

Communication from the Examining Division for Application No. EP06734083.6 mailed on Nov. 12, 2010, 3 pages. cited by applicant

Communication from the Examining Division for Application No. EP06734083.6 mailed on Oct. 23, 2009, 1 page. cited by applicant

Communication from the Examining Division for Application No. EP08746822.9 mailed on Jul. 13, 2010, 1 page. cited by applicant

Co-pending U.S. Appl. No. 61/286,283, filed Dec. 14, 2009. cited by applicant

Co-pending U.S. Appl. No. 61/297,462, filed Jan. 22, 2010. cited by applicant

Cox J.L., "Cardiac Surgery for Arrhythmias," Journal of Cardiovascular Electrophysiology, 2004, vol. 15, pp. 250-262. cited by applicant

Cox J.L., et al., "Five-Year Experience With the Maze Procedure for Atrial Fibrillation," The Annals of Thoracic Surgery, 1993, vol. 56, pp. 814-824. cited by applicant

Cox J.L., et al., "Modification of the Maze Procedure for Atrial Flutter and Atrial Fibrillation," The Journal of Thoracic and Cardiovascular Surgery, 1995, vol. 110, pp. 473-484. cited by applicant

Cox J.L., "The Status of Surgery for Cardiac Arrhythmias," Circulation, 1985, vol. 71, pp. 413-417. cited by applicant

Cox J.L., "The Surgical Treatment of Atrial Fibrillation," The Journal of Thoracic and Cardiovascular Surgery, 1991, vol. 101, pp. 584-592. cited by applicant

Elvan A., et al., "Radiofrequency Catheter Ablation of the Atria Reduces Inducibility and Duration of Atrial Fibrillation in Dogs," Circulation, vol. 91, 1995, pp. 2235-2244 [online], [retrieved Feb. 4, 2013]. Retrieved from the Internet:< URL: <http://circ.ahajournals.org/cgi/content/full/91/8/2235>>. cited by applicant

Elvan A., et al., "Radiofrequency Catheter Ablation (RFCA) of the Atria Effectively Abolishes Pacing Induced Chronic Atrial Fibrillation," Pacing and Clinical Electrophysiology, 1995, vol. 18, pp. 856. cited by applicant

Elvan, et al., "Replication of the 'Maze' Procedure by Radiofrequency Catheter Ablation Reduces the Ability to Induce Atrial Fibrillation," Pacing and Clinical Electrophysiology, 1994, vol. 17, pp. 774. cited by applicant

European Search Report for Application No. EP07799466.3 mailed on Nov. 18, 2010, 9 pages. cited by applicant

European Search Report for Application No. EP08746822.9 mailed on Mar. 29, 2010, 7 Pages. cited by applicant

Examination Communication for Application No. EP06734083.6 mailed on May 18, 2010, 3 Pages. cited by applicant

Extended European Search Report for Application No. EP06734083.6 mailed on Jul. 1, 2009, 6 pages. cited by applicant

Extended European search report for Application No. EP20070758716 mailed on Feb. 28, 2011, 8 Pages. cited by applicant

Extended European search report for Application No. EP20070799466 mailed on Nov. 18, 2010, 9 Pages. cited by applicant

Fieguth H.G., et al., "Inhibition of Atrial Fibrillation by Pulmonary Vein Isolation and Auricular Resection—Experimental Study in a Sheep Model," The European Journal of Cardio-Thoracic Surgery, 1997, vol. 11, pp. 714-721. cited by applicant

Final Office Action mailed Mar. 1, 2010 for U.S. Appl. No. 12/117,655, filed May 8, 2008. cited by applicant

Final Office Action mailed Jun. 2, 2011 for U.S. Appl. No. 12/117,655, filed May 8, 2008. cited by applicant

Final Office Action mailed Oct. 5, 2010 for U.S. Appl. No. 11/810,850, filed Jun. 7, 2007. cited by applicant

Final Office Action mailed May 12, 2011 for U.S. Appl. No. 11/775,771, filed Jul. 10, 2007. cited by applicant

Final Office Action mailed Sep. 16, 2010 for U.S. Appl. No. 11/828,267, filed Jul. 25, 2007. cited by applicant

Hoey M.F., et al., "Intramural Ablation Using Radiofrequency Energy Via Screw-Tip Catheter and Saline Electrode," Pacing and Clinical Electrophysiology, 1995, vol. 18, Part II, 487. cited by applicant

Huang, "Increase in the Lesion Size and Decrease in the Impedance Rise with a Saline Infusion Electrode Catheter for Radiofrequency," Circulation, 1989, vol. 80 (4), II-324. cited by applicant

International Search Report and Written Opinion for Application No. PCT/US2007/073184, mailed on Aug. 12, 2012, 7 pages. cited by applicant

International Search Report for Application No. PCT/US2006/003288, mailed on Aug. 9, 2007, 1 page. cited by applicant

International Search Report for Application No. PCT/US2007/064195, mailed on Dec. 7, 2007, 1 page. cited by applicant

International Search Report for Application No. PCT/US2007/071226, mailed on Sep. 4, 2008, 1 page. cited by applicant

International Search Report for Application No. PCT/US2007/077429, mailed on Apr. 7, 2008, 1 page. cited by applicant

Moser K.M., et al., "Angioscopic Visualization of Pulmonary Emboli," Chest, 1980, vol. 77 (2), pp. 198-201. cited by applicant

Nakamura F., et al., "Percutaneous Intracardiac Surgery With Cardioscopic Guidance," SPIE, 1992, vol. 1642, pp. 214-216. cited by applicant

Non-Final Office Action mailed Jun. 7, 2011 for U.S. Appl. No. 12/323,281, filed Nov. 25, 2008.
cited by applicant

Non-Final Office Action mailed Aug. 8, 2011 for U.S. Appl. No. 12/464,800, filed May 12, 2009.
cited by applicant

Non-Final Office Action mailed Jun. 8, 2009 for U.S. Appl. No. 12/117,655, filed May 8, 2008.
cited by applicant

Non-Final Office Action mailed May 9, 2011 for U.S. Appl. No. 11/961,950, filed Dec. 20, 2007.
cited by applicant

Non-Final Office Action mailed May 9, 2011 for U.S. Appl. No. 11/961,995, filed Dec. 20, 2007.
cited by applicant

Non-Final Office Action mailed May 9, 2011 for U.S. Appl. No. 11/962,029, filed Dec. 20, 2007.
cited by applicant

Non-Final Office Action mailed Jun. 10, 2010 for U.S. Appl. No. 11/560,742, filed Nov. 16, 2006.
cited by applicant

Non-Final Office Action mailed Apr. 11, 2011 for U.S. Appl. No. 11/763,399, filed Jun. 14, 2007.
cited by applicant

Non-Final Office Action mailed Mar. 11, 2011 for U.S. Appl. No. 11/848,202, filed Aug. 30, 2007.
cited by applicant

Non-Final Office Action mailed May 11, 2011 for U.S. Appl. No. 11/828,267, filed Jul. 25, 2007.
cited by applicant

Non-Final Office Action mailed Apr. 12, 2011 for U.S. Appl. No. 12/499,011, filed Jul. 7, 2009.
cited by applicant

Non-Final Office Action mailed Jan. 14, 2010 for U.S. Appl. No. 11/828,267, filed Jul. 25, 2007.
cited by applicant

Non-Final Office Action mailed Dec. 16, 2010 for U.S. Appl. No. 12/117,655, filed May 8, 2008.
cited by applicant

Non-Final Office Action mailed Mar. 16, 2010 for U.S. Appl. No. 11/810,850, filed Jun. 7, 2007.
cited by applicant

Non-Final Office Action mailed Feb. 18, 2011 for U.S. Appl. No. 12/947,198, filed Nov. 16, 2010.
cited by applicant

Non-Final Office Action mailed Feb. 18, 2011 for U.S. Appl. No. 12/947,246, filed Nov. 16, 2006.
cited by applicant

Non-Final Office Action mailed May 20, 2011 for U.S. Appl. No. 11/775,819, filed Jul. 10, 2007.
cited by applicant

Non-Final Office Action mailed May 20, 2011 for U.S. Appl. No. 11/877,386, filed Oct. 23, 2007.
cited by applicant

Non-Final Office Action mailed Jul. 21, 2010 for U.S. Appl. No. 11/687,597, filed Mar. 16, 2007.
cited by applicant

Non-Final Office Action mailed Apr. 22, 2011 for U.S. Appl. No. 12/367,019, filed Feb. 6, 2009.
cited by applicant

Non-Final Office Action mailed May 23, 2011 for U.S. Appl. No. 11/775,837, filed Jul. 10, 2007.
cited by applicant

Non-Final Office Action mailed Nov. 24, 2010 for U.S. Appl. No. 11/848,429, filed Aug. 31, 2007.
cited by applicant

Non-Final Office Action mailed Nov. 24, 2010 for U.S. Appl. No. 12/464,800, filed May 12, 2009.
cited by applicant

Non-Final Office Action mailed Apr. 25, 2011 for U.S. Appl. No. 11/959,158, filed Dec. 18, 2007.
cited by applicant

Non-Final Office Action mailed Feb. 25, 2010 for U.S. Appl. No. 11/259,498, filed Oct. 25, 2005.
cited by applicant

Non-Final Office Action mailed Feb. 25, 2011 for U.S. Appl. No. 11/848,207, filed Aug. 30, 2007. cited by applicant

Non-Final Office Action mailed Apr. 26, 2011 for U.S. Appl. No. 11/848,532, filed Aug. 31, 2007. cited by applicant

Non-Final Office Action mailed Apr. 27, 2011 for U.S. Appl. No. 11/828,281, filed Jul. 25, 2007. cited by applicant

Non-Final Office Action mailed Aug. 27, 2010 for U.S. Appl. No. 11/775,771, filed Jul. 10, 2007. cited by applicant

Non-Final Office Action mailed Dec. 27, 2010 for U.S. Appl. No. 12/026,455, filed Feb. 5, 2008. cited by applicant

Notice of Allowance mailed Feb. 3, 2011 for U.S. Appl. No. 11/560,732, filed Nov. 16, 2006. cited by applicant

Notice of Allowance mailed Jun. 13, 2011 for Japanese Application No. 2007-554156 filed Jan. 30, 2006. cited by applicant

Notice of Allowance mailed Nov. 15, 2010 for U.S. Appl. No. 11/259,498, filed Oct. 25, 2005. cited by applicant

Notice of Allowance mailed Nov. 15, 2010 for U.S. Appl. No. 11/560,742, filed Nov. 16, 2006. cited by applicant

Notice of Allowance mailed Feb. 24, 2011 for U.S. Appl. No. 11/560,732, filed Mar. 16, 2007. cited by applicant

Notice of Allowance mailed Feb. 24, 2011 for U.S. Appl. No. 11/687,597, filed Mar. 16, 2007. cited by applicant

Office Action mailed Feb. 15, 2011 for Japanese Application No. 2007-554156 filed Jan. 30, 2006. cited by applicant

Office Action mailed Apr. 27, 2011 for Japanese Application No. 2009-500630 filed Mar. 16, 2007. cited by applicant

Pappone C., et al., "Circumferential Radiofrequency Ablation of Pulmonary Vein Ostia," Circulation, 2000, vol. 102, pp. 2619-2628. cited by applicant

Sethi K.K., et al., "Transseptal catheterization for the electrophysiologist: modification with a 'view'," Journal of Interventional Cardiac Electrophysiology, 2001, vol. 5 (1), pp. 97-99. cited by applicant

Tse HF., et al., "Angiogenesis in Ischaemic Myocardium by Intramyocardial Autologous Bone Marrow Mononuclear Cell Implantation," Lancet, 2003, vol. 361, pp. 47-49. cited by applicant

Supplemental European Search Report for Application No. EP07758716 mailed on Feb. 28, 2011, 8 Pages. cited by applicant

Supplementary European search report for Application No. EP07812146.4 mailed on Nov. 18, 2010, 8 Pages. cited by applicant

Supplementary European Search Report for Application No. EP07841754, mailed on Jun. 30, 2010, 6 pages. cited by applicant

Thiagalingam A., et al., "Cooled Needle Catheter Ablation Creates Deeper and Wider Lesions than Irrigated Tip Catheter Ablation," Journal of Cardiovascular Electrophysiology, 2005, vol. 16 (5), pp. 1-8. cited by applicant

Uchida Y., "Developmental History of Cardioscopes", in: Coronary Angioscopy, Chapter 19, Futura Publishing Company, Inc., 2001, pp. 187-197. cited by applicant

Willkampf F.H., et al., "Radiofrequency Ablation with a Cooled Porous Electrode Catheter," JACC, Abstract, 1988, vol. 11 (2), pp. 17A. cited by applicant

Written Opinion for Application No. PCT/US2006/003288, mailed on Aug. 9, 2007, 6 pages. cited by applicant

Written Opinion for Application No. PCT/US2007/064195, mailed on Dec. 7, 2007, 5 pages. cited by applicant

Written Opinion for Application No. PCT/US2007/071226, mailed on Sep. 4, 2008, 4 page. cited by applicant
Written Opinion for Application No. PCT/US2007/077429, mailed on Apr. 7, 2008, 5 pages. cited by applicant
Vertut, J, and Coiffet, P., "Robot Technology: Teleoperation and Robotics Evolution and Development," English translation, Prentice-Hall, Inc., Englewood Cliffs, NJ, USA 1986, vol. 3A, 332 pages. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 15/999,543 filed Aug. 20, 2018 which is a continuation of U.S. patent application Ser. No. 14/662,960 filed Mar. 19, 2015 which is a continuation of U.S. patent application Ser. No. 12/947,198 filed Nov. 16, 2010, which is a continuation of U.S. patent application Ser. No. 11/259,498 filed Oct. 25, 2005, which claims the benefit of priority to U.S. Provisional Patent Application No. 60/649,246 filed Feb. 2, 2005, the disclosures of which are hereby incorporated by reference in their entirety.

FIELD OF THE INVENTION

- (1) The present invention relates generally to medical devices used for visualizing and/or manipulating regions of tissue within a body. More particularly, the present invention relates to apparatus and methods for visualizing and/or manipulating tissue regions within a body lumen, e.g., tissue surrounding or adjacent to valves within a heart, which are generally difficult to image because of surrounding opaque bodily fluids such as blood or the tissue of the inter-atrial septum for trans-septal procedures.
- (2) Conventional devices for visualizing interior regions of a body lumen are known. For example, ultrasound devices have been used to produce images from within a body in vivo. Ultrasound has been used both with and without contrast agents, which typically enhance ultrasound-derived images.
- (3) Other conventional methods have utilized catheters or probes having position sensors deployed within the body lumen, such as the interior of a cardiac chamber. These types of positional sensors are typically used to determine the movement of a cardiac tissue surface or the electrical activity within the cardiac tissue. When a sufficient number of points have been sampled by the sensors, a "map" of the cardiac tissue may be generated.
- (4) Another conventional device utilizes an inflatable balloon which is typically introduced intravascularly in a deflated state and then inflated against the tissue region to be examined. Imaging is typically accomplished by an optical fiber or other apparatus such as electronic chips for viewing the tissue through the membrane(s) of the inflated balloon. Moreover, the balloon must generally be inflated for imaging. Other conventional balloons utilize a cavity or depression formed at a distal end of the inflated balloon. This cavity or depression is pressed against the tissue to be examined and is flushed with a clear fluid to provide a clear pathway through the blood.
- (5) However, such imaging balloons have many inherent disadvantages. For instance, such balloons generally require that the balloon be inflated to a relatively large size which may undesirably displace surrounding tissue and interfere with fine positioning of the imaging system against the tissue. Moreover, the working area created by such inflatable balloons are generally cramped and

limited in size. Furthermore, inflated balloons may be susceptible to pressure changes in the surrounding fluid. For example, if the environment surrounding the inflated balloon undergoes pressure changes, e.g., during systolic and diastolic pressure cycles in a beating heart, the constant pressure change may affect the inflated balloon volume and its positioning to produce unsteady or undesirable conditions for optimal tissue imaging.

(6) Accordingly, these types of imaging modalities are generally unable to provide desirable images useful for sufficient diagnosis and therapy of the endoluminal structure, due in part to factors such as dynamic forces generated by the natural movement of the heart. Moreover, anatomic structures within the body can occlude or obstruct the image acquisition process. Also, the presence and movement of opaque bodily fluids such as blood generally make in vivo imaging of tissue regions within the heart difficult.

(7) Other external imaging modalities are also conventionally utilized. For example, computed tomography (CT) and magnetic resonance imaging (MRI) are typical modalities which are widely used to obtain images of body lumens such as the interior chambers of the heart. However, such imaging modalities fail to provide real-time imaging for intra-operative therapeutic procedures. Fluoroscopic imaging, for instance, is widely used to identify anatomic landmarks within the heart and other regions of the body. However, fluoroscopy fails to provide an accurate image of the tissue quality or surface and also fails to provide for instrumentation for performing tissue manipulation or other therapeutic procedures upon the visualized tissue regions. In addition, fluoroscopy provides a shadow of the intervening tissue onto a plate or sensor when it may be desirable to view the intraluminal surface of the tissue to diagnose pathologies or to perform some form of therapy on it.

(8) Thus, a tissue imaging system which is able to provide real-time in vivo images of tissue regions within body lumens such as the heart through opaque media such as blood and which also provide instruments for therapeutic procedures upon the visualized tissue are desirable.

BRIEF SUMMARY OF THE INVENTION

(9) A tissue imaging and manipulation apparatus that may be utilized for procedures within a body lumen, such as the heart, in which visualization of the surrounding tissue is made difficult, if not impossible, by medium contained within the lumen such as blood, is described below. Generally, such a tissue imaging and manipulation apparatus comprises an optional delivery catheter or sheath through which a deployment catheter and imaging hood may be advanced for placement against or adjacent to the tissue to be imaged.

(10) The deployment catheter may define a fluid delivery lumen therethrough as well as an imaging lumen within which an optical imaging fiber or assembly may be disposed for imaging tissue. When deployed, the imaging hood may be expanded into any number of shapes, e.g., cylindrical, conical as shown, semi-spherical, etc., provided that an open area or field is defined by the imaging hood. The open area is the area within which the tissue region of interest may be imaged. The imaging hood may also define an atraumatic contact lip or edge for placement or abutment against the tissue region of interest. Moreover, the distal end of the deployment catheter or separate manipulatable catheters may be articulated through various controlling mechanisms such as push-pull wires manually or via computer control.

(11) The deployment catheter may also be stabilized relative to the tissue surface through various methods. For instance, inflatable stabilizing balloons positioned along a length of the catheter may be utilized, or tissue engagement anchors may be passed through or along the deployment catheter for temporary engagement of the underlying tissue.

(12) In operation, after the imaging hood has been deployed, fluid may be pumped at a positive pressure through the fluid delivery lumen until the fluid fills the open area completely and displaces any blood from within the open area. The fluid may comprise any biocompatible fluid, e.g., saline, water, plasma, Fluorinert™, etc., which is sufficiently transparent to allow for relatively undistorted visualization through the fluid. The fluid may be pumped continuously or

intermittently to allow for image capture by an optional processor which may be in communication with the assembly.

(13) The imaging hood may be formed into any number of configurations and the imaging assembly may also be utilized with any number of therapeutic tools which may be deployed through the deployment catheter.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1A shows a side view of one variation of a tissue imaging apparatus during deployment from a sheath or delivery catheter.

(2) FIG. 1B shows the deployed tissue imaging apparatus of FIG. 1A having an optionally expandable hood or sheath attached to an imaging and/or diagnostic catheter.

(3) FIG. 1C shows an end view of a deployed imaging apparatus.

(4) FIGS. 1D to 1F show the apparatus of FIGS. 1A to 1C with an additional lumen, e.g., for passage of a guidewire therethrough.

(5) FIGS. 2A and 2B show one example of a deployed tissue imager positioned against or adjacent to the tissue to be imaged and a flow of fluid, such as saline, displacing blood from within the expandable hood.

(6) FIG. 3A shows an articulatable imaging assembly which may be manipulated via push-pull wires or by computer control.

(7) FIGS. 3B and 3C show steerable instruments, respectively, where an articulatable delivery catheter may be steered within the imaging hood or a distal portion of the deployment catheter itself may be steered.

(8) FIGS. 4A to 4C show side and cross-sectional end views, respectively, of another variation having an off-axis imaging capability.

(9) FIG. 5 shows an illustrative view of an example of a tissue imager advanced intravascularly within a heart for imaging tissue regions within an atrial chamber.

(10) FIGS. 6A to 6C illustrate deployment catheters having one or more optional inflatable balloons or anchors for stabilizing the device during a procedure.

(11) FIGS. 7A and 7B illustrate a variation of an anchoring mechanism such as a helical tissue piercing device for temporarily stabilizing the imaging hood relative to a tissue surface.

(12) FIG. 7C shows another variation for anchoring the imaging hood having one or more tubular support members integrated with the imaging hood, each support members may define a lumen therethrough for advancing a helical tissue anchor within.

(13) FIG. 8A shows an illustrative example of one variation of how a tissue imager may be utilized with an imaging device.

(14) FIG. 8B shows a further illustration of a hand-held variation of the fluid delivery and tissue manipulation system.

(15) FIGS. 9A to 9C illustrate an example of capturing several images of the tissue at multiple regions.

(16) FIGS. 10A and 10B show charts illustrating how fluid pressure within the imaging hood may be coordinated with the surrounding blood pressure; the fluid pressure in the imaging hood may be coordinated with the blood pressure or it may be regulated based upon pressure feedback from the blood.

(17) FIG. 11A shows a side view of another variation of a tissue imager having an imaging balloon within an expandable hood.

(18) FIG. 11B shows another variation of a tissue imager utilizing a translucent or transparent imaging balloon.

(19) FIG. 12A shows another variation in which a flexible expandable or distensible membrane may be incorporated within the imaging hood to alter the volume of fluid dispensed.

(20) FIGS. 12B and 12C show another variation in which the imaging hood may be partially or selectively deployed from the catheter to alter the area of the tissue being visualized as well as the volume of the dispensed fluid.

(21) FIGS. 13A and 13B show exemplary side and cross-sectional views, respectively, of another variation in which the injected fluid may be drawn back into the device for minimizing fluid input into a body being treated.

(22) FIGS. 14A to 14D show various configurations and methods for configuring an imaging hood into a low-profile for delivery and/or deployment.

(23) FIGS. 15A and 15B show an imaging hood having an helically expanding frame or support.

(24) FIGS. 16A and 16B show another imaging hood having one or more hood support members, which are pivotably attached at their proximal ends to deployment catheter, integrated with a hood membrane.

(25) FIGS. 17A and 17B show yet another variation of the imaging hood having at least two or more longitudinally positioned support members supporting the imaging hood membrane where the support members are movable relative to one another via a torquing or pulling or pushing force.

(26) FIGS. 18A and 18B show another variation where a distal portion of the deployment catheter may have several pivoting members which form a tubular shape in its low profile configuration.

(27) FIGS. 19A and 19B show another variation where the distal portion of deployment catheter may be fabricated from a flexible metallic or polymeric material to form a radially expanding hood.

(28) FIGS. 20A and 20B show another variation where the imaging hood may be formed from a plurality of overlapping hood members which overlies one another in an overlapping pattern.

(29) FIGS. 21A and 21B show another example of an expandable hood which is highly conformable against tissue anatomy with varying geography.

(30) FIG. 22A shows yet another example of an expandable hood having a number of optional electrodes placed about the contact edge or lip of the hood for sensing tissue contact or detecting arrhythmias.

(31) FIG. 22B shows another variation for conforming the imaging hood against the underlying tissue where an inflatable contact edge may be disposed around the circumference of the imaging hood.

(32) FIG. 23 shows a variation of the system which may be instrumented with a transducer for detecting the presence of blood seeping back into the imaging hood.

(33) FIGS. 24A and 24B show variations of the imaging hood instrumented with sensors for detecting various physical parameters; the sensors may be instrumented around the outer surface of the imaging hood and also within the imaging hood.

(34) FIGS. 25A and 25B show a variation where the imaging hood may have one or more LEDs over the hood itself for providing illumination of the tissue to be visualized.

(35) FIGS. 26A and 26B show another variation in which a separate illumination tool having one or more LEDs mounted thereon may be utilized within the imaging hood.

(36) FIG. 27 shows one example of how a therapeutic tool may be advanced through the tissue imager for treating a tissue region of interest.

(37) FIG. 28 shows another example of a helical therapeutic tool for treating the tissue region of interest.

(38) FIG. 29 shows a variation of how a therapeutic tool may be utilized with an expandable imaging balloon.

(39) FIGS. 30A and 30B show alternative configurations for therapeutic instruments which may be utilized; one variation is shown having an angled instrument arm and another variation is shown with an off-axis instrument arm.

(40) FIGS. 31A to 31C show side and end views, respectively, of an imaging system which may be

utilized with an ablation probe.

(41) FIGS. **32A** and **32B** show side and end views, respectively, of another variation of the imaging hood with an ablation probe, where the imaging hood may be enclosed for regulating a temperature of the underlying tissue.

(42) FIGS. **33A** and **33B** show an example in which the imaging fluid itself may be altered in temperature to facilitate various procedures upon the underlying tissue.

(43) FIGS. **34A** and **34B** show an example of a laser ring generator which may be utilized with the imaging system and an example for applying the laser ring generator within the left atrium of a heart for treating atrial fibrillation.

(44) FIGS. **35A** to **35C** show an example of an extendible cannula generally comprising an elongate tubular member which may be positioned within the deployment catheter during delivery and then projected distally through the imaging hood and optionally beyond.

(45) FIGS. **36A** and **36B** show side and end views, respectively, of an imaging hood having one or more tubular support members integrated with the hood for passing instruments or tools therethrough for treatment upon the underlying tissue.

(46) FIGS. **37A** and **37B** illustrate how an imaging device may be guided within a heart chamber to a region of interest utilizing a lighted probe positioned temporarily within, e.g., a lumen of the coronary sinus.

(47) FIGS. **38A** and **38B** show an imaging hood having a removable disk-shaped member for implantation upon the tissue surface.

(48) FIGS. **39A** to **39C** show one method for implanting the removable disk of FIGS. **38A** and **38B**.

(49) FIGS. **40A** and **40B** illustrate an imaging hood having a deployable anchor assembly attached to the tissue contact edge and an assembly view of the anchors and the suture or wire connected to the anchors, respectively.

(50) FIGS. **41A** to **41D** show one method for deploying the anchor assembly of FIGS. **40A** and **40B** for closing an opening or wound.

(51) FIG. **42** shows another variation in which the imaging system may be fluidly coupled to a dialysis unit for filtering a patient's blood.

(52) FIGS. **43A** and **43B** show a variation of the deployment catheter having a first deployable hood and a second deployable hood positioned distal to the first hood; the deployment catheter may also have a side-viewing imaging element positioned between the first and second hoods for imaging tissue between the expanded hoods.

(53) FIGS. **44A** and **44B** show side and end views, respectively, of a deployment catheter having a side-imaging balloon in an un-inflated low-profile configuration.

(54) FIGS. **45A** to **45C** show side, top, and end views, respectively, of the inflated balloon of FIGS. **44A** and **44B** defining a visualization field in the inflated balloon.

(55) FIGS. **46A** and **46B** show side and cross-sectional end views, respectively, for one method of use in visualizing a lesion upon a vessel wall within the visualization field of the inflated balloon from FIGS. **45A** to **45C**.

DETAILED DESCRIPTION OF THE INVENTION

(56) A tissue-imaging and manipulation apparatus described below is able to provide real-time images in vivo of tissue regions within a body lumen such as a heart, which is filled with blood flowing dynamically therethrough and is also able to provide intravascular tools and instruments for performing various procedures upon the imaged tissue regions. Such an apparatus may be utilized for many procedures, e.g., facilitating trans-septal access to the left atrium, cannulating the coronary sinus, diagnosis of valve regurgitation/stenosis, valvuloplasty, atrial appendage closure, arrhythmogenic focus ablation, among other procedures.

(57) One variation of a tissue access and imaging apparatus is shown in the detail perspective views of FIGS. **1A** to **1C**. As shown in FIG. **1A**, tissue imaging and manipulation assembly **10** may be delivered intravascularly through the patient's body in a low-profile configuration via a delivery

catheter or sheath **14**. In the case of treating tissue, such as the mitral valve located at the outflow tract of the left atrium of the heart, it is generally desirable to enter or access the left atrium while minimizing trauma to the patient. To non-operatively effect such access, one conventional approach involves puncturing the intra-atrial septum from the right atrial chamber to the left atrial chamber in a procedure commonly called a trans-septal procedure or septostomy. For procedures such as percutaneous valve repair and replacement, trans-septal access to the left atrial chamber of the heart may allow for larger devices to be introduced into the venous system than can generally be introduced percutaneously into the arterial system.

(58) When the imaging and manipulation assembly **10** is ready to be utilized for imaging tissue, imaging hood **12** may be advanced relative to catheter **14** and deployed from a distal opening of catheter **14**, as shown by the arrow. Upon deployment, imaging hood **12** may be unconstrained to expand or open into a deployed imaging configuration, as shown in FIG. **1B**. Imaging hood **12** may be fabricated from a variety of pliable or conformable biocompatible material including but not limited to, e.g., polymeric, plastic, or woven materials. One example of a woven material is Kevlar® (E. I. du Pont de Nemours, Wilmington, Del.), which is an aramid and which can be made into thin, e.g., less than 0.001 in., materials which maintain enough integrity for such applications described herein. Moreover, the imaging hood **12** may be fabricated from a translucent or opaque material and in a variety of different colors to optimize or attenuate any reflected lighting from surrounding fluids or structures, i.e., anatomical or mechanical structures or instruments. In either case, imaging hood **12** may be fabricated into a uniform structure or a scaffold-supported structure, in which case a scaffold made of a shape memory alloy, such as Nitinol, or a spring steel, or plastic, etc., may be fabricated and covered with the polymeric, plastic, or woven material.

(59) Imaging hood **12** may be attached at interface **24** to a deployment catheter **16** which may be translated independently of deployment catheter or sheath **14**. Attachment of interface **24** may be accomplished through any number of conventional methods. Deployment catheter **16** may define a fluid delivery lumen **18** as well as an imaging lumen **20** within which an optical imaging fiber or assembly may be disposed for imaging tissue. When deployed, imaging hood **12** may expand into any number of shapes, e.g., cylindrical, conical as shown, semi-spherical, etc., provided that an open area or field **26** is defined by imaging hood **12**. The open area **26** is the area within which the tissue region of interest may be imaged. Imaging hood **12** may also define an atraumatic contact lip or edge **22** for placement or abutment against the tissue region of interest. Moreover, the diameter of imaging hood **12** at its maximum fully deployed diameter, e.g., at contact lip or edge **22**, is typically greater relative to a diameter of the deployment catheter **16** (although a diameter of contact lip or edge **22** may be made to have a smaller or equal diameter of deployment catheter **16**). For instance, the contact edge diameter may range anywhere from 1 to 5 times (or even greater, as practicable) a diameter of deployment catheter **16**. FIG. **1C** shows an end view of the imaging hood **12** in its deployed configuration. Also shown are the contact lip or edge **22** and fluid delivery lumen **18** and imaging lumen **20**.

(60) The imaging and manipulation assembly **10** may additionally define a guidewire lumen therethrough, e.g., a concentric or eccentric lumen, as shown in the side and end views, respectively, of FIGS. **1D** to **1F**. The deployment catheter **16** may define guidewire lumen **19** for facilitating the passage of the system over or along a guidewire **17**, which may be advanced intravascularly within a body lumen. The deployment catheter **16** may then be advanced over the guidewire **17**, as generally known in the art.

(61) During operation, after imaging hood **12** has been deployed, as in FIG. **1B**, and desirably positioned against the tissue region to be imaged along contact edge **22**, the displacing fluid may be pumped at positive pressure through fluid delivery lumen **18** until the fluid fills open area **26** completely and displaces any fluid **28** from within open area **26**. The displacing fluid flow may be laminarized to improve its clearing effect and to help prevent blood from re-entering the imaging hood **12**. Alternatively, fluid flow may be started before the deployment takes place. The displacing

fluid, also described herein as imaging fluid, may comprise any biocompatible fluid, e.g., saline, water, plasma, etc., which is sufficiently transparent to allow for relatively undistorted visualization through the fluid. Alternatively or additionally, any number of therapeutic drugs may be suspended within the fluid or may comprise the fluid itself which is pumped into open area **26** and which is subsequently passed into and through the heart and the patient body.

(62) As seen in the example of FIGS. **2A** and **2B**, deployment catheter **16** may be manipulated to position deployed imaging hood **12** against or near the underlying tissue region of interest to be imaged, in this example a portion of annulus **A** of mitral valve **MV** within the left atrial chamber. As the surrounding blood **30** flows around imaging hood **12** and within open area **26** defined within imaging hood **12**, as seen in FIG. **2A**, the underlying annulus **A** is obstructed by the opaque blood **30** and is difficult to view through the imaging lumen **20**. The translucent fluid **28**, such as saline, may then be pumped through fluid delivery lumen **18**, intermittently or continuously, until the blood **30** is at least partially, and preferably completely, displaced from within open area **26** by fluid **28**, as shown in FIG. **2B**.

(63) Although contact edge **22** need not directly contact the underlying tissue, it is at least preferably brought into close proximity to the tissue such that the flow of clear fluid **28** from open area **26** may be maintained to inhibit significant backflow of blood **30** back into open area **26**. Contact edge **22** may also be made of a soft elastomeric material such as certain soft grades of silicone or polyurethane, as typically known, to help contact edge **22** conform to an uneven or rough underlying anatomical tissue surface. Once the blood **30** has been displaced from imaging hood **12**, an image may then be viewed of the underlying tissue through the clear fluid **30**. This image may then be recorded or available for real-time viewing for performing a therapeutic procedure. The positive flow of fluid **28** may be maintained continuously to provide for clear viewing of the underlying tissue. Alternatively, the fluid **28** may be pumped temporarily or sporadically only until a clear view of the tissue is available to be imaged and recorded, at which point the fluid flow **28** may cease and blood **30** may be allowed to seep or flow back into imaging hood **12**. This process may be repeated a number of times at the same tissue region or at multiple tissue regions.

(64) In desirably positioning the assembly at various regions within the patient body, a number of articulation and manipulation controls may be utilized. For example, as shown in the articulatable imaging assembly **40** in FIG. **3A**, one or more push-pull wires **42** may be routed through deployment catheter **16** for steering the distal end portion of the device in various directions **46** to desirably position the imaging hood **12** adjacent to a region of tissue to be visualized. Depending upon the positioning and the number of push-pull wires **42** utilized, deployment catheter **16** and imaging hood **12** may be articulated into any number of configurations **44**. The push-pull wire or wires **42** may be articulated via their proximal ends from outside the patient body manually utilizing one or more controls. Alternatively, deployment catheter **16** may be articulated by computer control, as further described below.

(65) Additionally or alternatively, an articulatable delivery catheter **48**, which may be articulated via one or more push-pull wires and having an imaging lumen and one or more working lumens, may be delivered through the deployment catheter **16** and into imaging hood **12**. With a distal portion of articulatable delivery catheter **48** within imaging hood **12**, the clear displacing fluid may be pumped through delivery catheter **48** or deployment catheter **16** to clear the field within imaging hood **12**. As shown in FIG. **3B**, the articulatable delivery catheter **48** may be articulated within the imaging hood to obtain a better image of tissue adjacent to the imaging hood **12**. Moreover, articulatable delivery catheter **48** may be articulated to direct an instrument or tool passed through the catheter **48**, as described in detail below, to specific areas of tissue imaged through imaging hood **12** without having to reposition deployment catheter **16** and re-clear the imaging field within hood **12**.

(66) Alternatively, rather than passing an articulatable delivery catheter **48** through the deployment

catheter **16**, a distal portion of the deployment catheter **16** itself may comprise a distal end **49** which is articulatable within imaging hood **12**, as shown in FIG. **3C**. Directed imaging, instrument delivery, etc., may be accomplished directly through one or more lumens within deployment catheter **16** to specific regions of the underlying tissue imaged within imaging hood **12**.

(67) Visualization within the imaging hood **12** may be accomplished through an imaging lumen **20** defined through deployment catheter **16**, as described above. In such a configuration, visualization is available in a straight-line manner, i.e., images are generated from the field distally along a longitudinal axis defined by the deployment catheter **16**. Alternatively or additionally, an articulatable imaging assembly having a pivotable support member **50** may be connected to, mounted to, or otherwise passed through deployment catheter **16** to provide for visualization off-axis relative to the longitudinal axis defined by deployment catheter **16**, as shown in FIG. **4A**. Support member **50** may have an imaging element **52**, e.g., a CCD or CMOS imager or optical fiber, attached at its distal end with its proximal end connected to deployment catheter **16** via a pivoting connection **54**.

(68) If one or more optical fibers are utilized for imaging, the optical fibers **58** may be passed through deployment catheter **16**, as shown in the cross-section of FIG. **4B**, and routed through the support member **50**. The use of optical fibers **58** may provide for increased diameter sizes of the one or several lumens **56** through deployment catheter **16** for the passage of diagnostic and/or therapeutic tools therethrough. Alternatively, electronic chips, such as a charge coupled device (CCD) or a CMOS imager, which are typically known, may be utilized in place of the optical fibers **58**, in which case the electronic imager may be positioned in the distal portion of the deployment catheter **16** with electric wires being routed proximally through the deployment catheter **16**. Alternatively, the electronic imagers may be wirelessly coupled to a receiver for the wireless transmission of images. Additional optical fibers or light emitting diodes (LEDs) can be used to provide lighting for the image or operative theater, as described below in further detail. Support member **50** may be pivoted via connection **54** such that the member **50** can be positioned in a low-profile configuration within channel or groove **60** defined in a distal portion of catheter **16**, as shown in the cross-section of FIG. **4C**. During intravascular delivery of deployment catheter **16** through the patient body, support member **50** can be positioned within channel or groove **60** with imaging hood **12** also in its low-profile configuration. During visualization, imaging hood **12** may be expanded into its deployed configuration and support member **50** may be deployed into its off-axis configuration for imaging the tissue adjacent to hood **12**, as in FIG. **4A**. Other configurations for support member **50** for off-axis visualization may be utilized, as desired.

(69) FIG. **5** shows an illustrative cross-sectional view of a heart **H** having tissue regions of interest being viewed via an imaging assembly **10**. In this example, delivery catheter assembly **70** may be introduced percutaneously into the patient's vasculature and advanced through the superior vena cava SVC and into the right atrium RA. The delivery catheter or sheath **72** may be articulated through the atrial septum AS and into the left atrium LA for viewing or treating the tissue, e.g., the annulus A, surrounding the mitral valve MV. As shown, deployment catheter **16** and imaging hood **12** may be advanced out of delivery catheter **72** and brought into contact or in proximity to the tissue region of interest. In other examples, delivery catheter assembly **70** may be advanced through the inferior vena cava IVC, if so desired. Moreover, other regions of the heart **H**, e.g., the right ventricle RV or left ventricle LV, may also be accessed and imaged or treated by imaging assembly **10**.

(70) In accessing regions of the heart **H** or other parts of the body, the delivery catheter or sheath **14** may comprise a conventional intra-vascular catheter or an endoluminal delivery device. Alternatively, robotically-controlled delivery catheters may also be optionally utilized with the imaging assembly described herein, in which case a computer-controller **74** may be used to control the articulation and positioning of the delivery catheter **14**. An example of a robotically-controlled delivery catheter which may be utilized is described in further detail in U.S. Pat. Pub.

2002/0087169 A1 to Brock et al. entitled “Flexible Instrument”, which is incorporated herein by reference in its entirety. Other robotically-controlled delivery catheters manufactured by Hansen Medical, Inc. (Mountain View, Calif.) may also be utilized with the delivery catheter **14**.

(71) To facilitate stabilization of the deployment catheter **16** during a procedure, one or more inflatable balloons or anchors **76** may be positioned along the length of catheter **16**, as shown in FIG. **6A**. For example, when utilizing a trans-septal approach across the atrial septum AS into the left atrium LA, the inflatable balloons **76** may be inflated from a low-profile into their expanded configuration to temporarily anchor or stabilize the catheter **16** position relative to the heart H. FIG. **6B** shows a first balloon **78** inflated while FIG. **6C** also shows a second balloon **80** inflated proximal to the first balloon **78**. In such a configuration, the septal wall AS may be wedged or sandwiched between the balloons **78**, **80** to temporarily stabilize the catheter **16** and imaging hood **12**. A single balloon **78** or both balloons **78**, **80** may be used. Other alternatives may utilize expandable mesh members, malecots, or any other temporary expandable structure. After a procedure has been accomplished, the balloon assembly **76** may be deflated or re-configured into a low-profile for removal of the deployment catheter **16**.

(72) To further stabilize a position of the imaging hood **12** relative to a tissue surface to be imaged, various anchoring mechanisms may be optionally employed for temporarily holding the imaging hood **12** against the tissue. Such anchoring mechanisms may be particularly useful for imaging tissue which is subject to movement, e.g., when imaging tissue within the chambers of a beating heart. A tool delivery catheter **82** having at least one instrument lumen and an optional visualization lumen may be delivered through deployment catheter **16** and into an expanded imaging hood **12**. As the imaging hood **12** is brought into contact against a tissue surface T to be examined, an anchoring mechanisms such as a helical tissue piercing device **84** may be passed through the tool delivery catheter **82**, as shown in FIG. **7A**, and into imaging hood **12**.

(73) The helical tissue engaging device **84** may be torqued from its proximal end outside the patient body to temporarily anchor itself into the underlying tissue surface T. Once embedded within the tissue T, the helical tissue engaging device **84** may be pulled proximally relative to deployment catheter **16** while the deployment catheter **16** and imaging hood **12** are pushed distally, as indicated by the arrows in FIG. **7B**, to gently force the contact edge or lip **22** of imaging hood against the tissue T. The positioning of the tissue engaging device **84** may be locked temporarily relative to the deployment catheter **16** to ensure secure positioning of the imaging hood **12** during a diagnostic or therapeutic procedure within the imaging hood **12**. After a procedure, tissue engaging device **84** may be disengaged from the tissue by torquing its proximal end in the opposite direction to remove the anchor from the tissue T and the deployment catheter **16** may be repositioned to another region of tissue where the anchoring process may be repeated or removed from the patient body. The tissue engaging device **84** may also be constructed from other known tissue engaging devices such as vacuum-assisted engagement or grasper-assisted engagement tools, among others.

(74) Although a helical anchor **84** is shown, this is intended to be illustrative and other types of temporary anchors may be utilized, e.g., hooked or barbed anchors, graspers, etc. Moreover, the tool delivery catheter **82** may be omitted entirely and the anchoring device may be delivered directly through a lumen defined through the deployment catheter **16**.

(75) In another variation where the tool delivery catheter **82** may be omitted entirely to temporarily anchor imaging hood **12**, FIG. **7C** shows an imaging hood **12** having one or more tubular support members **86**, e.g. four support members **86** as shown, integrated with the imaging hood **12**. The tubular support members **86** may define lumens therethrough each having helical tissue engaging devices **88** positioned within. When an expanded imaging hood **12** is to be temporarily anchored to the tissue, the helical tissue engaging devices **88** may be urged distally to extend from imaging hood **12** and each may be torqued from its proximal end to engage the underlying tissue T. Each of the helical tissue engaging devices **88** may be advanced through the length of deployment catheter **16** or they may be positioned within tubular support members **86** during the delivery and

deployment of imaging hood **12**. Once the procedure within-imaging hood **12** is finished, each of the tissue engaging devices **88** may be disengaged from the tissue and the imaging hood **12** may be repositioned to another region of tissue or removed from the patient body.

(76) An illustrative example is shown in FIG. **8A** of a tissue imaging assembly connected to a fluid delivery system **90** and to an optional processor **98** and image recorder and/or viewer **100**. The fluid delivery system **90** may generally comprise a pump **92** and an optional valve **94** for controlling the flow rate of the fluid into the system. A fluid reservoir **96**, fluidly connected to pump **92**, may hold the fluid to be pumped through imaging hood **12**. An optional central processing unit or processor **98** may be in electrical communication with fluid delivery system **90** for controlling flow parameters such as the flow rate and/or velocity of the pumped fluid. The processor **98** may also be in electrical communication with an image recorder and/or viewer **100** for directly viewing the images of tissue received from within imaging hood **12**. Imager recorder and/or viewer **100** may also be used not only to record the image but also the location of the viewed tissue region, if so desired.

(77) Optionally, processor **98** may also be utilized to coordinate the fluid flow and the image capture. For instance, processor **98** may be programmed to provide for fluid flow from reservoir **96** until the tissue area has been displaced of blood to obtain a clear image. Once the image has been determined to be sufficiently clear, either visually by a practitioner or by computer, an image of the tissue may be captured automatically by recorder **100** and pump **92** may be automatically stopped or slowed by processor **98** to cease the fluid flow into the patient. Other variations for fluid delivery and image capture are, of course, possible and the aforementioned configuration is intended only to be illustrative and not limiting.

(78) FIG. **8B** shows a further illustration of a hand-held variation of the fluid delivery and tissue manipulation system **110**. In this variation, system **110** may have a housing or handle assembly **112** which can be held or manipulated by the physician from outside the patient body. The fluid reservoir **114**, shown in this variation as a syringe, can be fluidly coupled to the handle assembly **112** and actuated via a pumping mechanism **116**, e.g., lead screw. Fluid reservoir **114** may be a simple reservoir separated from the handle assembly **112** and fluidly coupled to handle assembly **112** via one or more tubes. The fluid flow rate and other mechanisms may be metered by the electronic controller **118**.

(79) Deployment of imaging hood **12** may be actuated by a hood deployment switch **120** located on the handle assembly **112** while dispensation of the fluid from reservoir **114** may be actuated by a fluid deployment switch **122**, which can be electrically coupled to the controller **118**. Controller **118** may also be electrically coupled to a wired or wireless antenna **124** optionally integrated with the handle assembly **112**, as shown in the figure. The wireless antenna **124** can be used to wirelessly transmit images captured from the imaging hood **12** to a receiver, e.g., via Bluetooth® wireless technology (Bluetooth SIG. Inc., Bellevue, Wash.), RF, etc., for viewing on a monitor **128** or for recording for later viewing.

(80) Articulation control of the deployment catheter **16**, or a delivery catheter or sheath **14** through which the deployment catheter **16** may be delivered, may be accomplished by computer control, as described above, in which case an additional controller may be utilized with handle assembly **112**. In the case of manual articulation, handle assembly **112** may incorporate one or more articulation controls **126** for manual manipulation of the position of deployment catheter **16**. Handle assembly **112** may also define one or more instrument ports **130** through which a number of intravascular tools may be passed for tissue manipulation and treatment within imaging hood **12**, as described further below. Furthermore, in certain procedures, fluid or debris may be sucked into imaging hood **12** for evacuation from the patient body by optionally fluidly coupling a suction pump **132** to handle assembly **112** or directly to deployment catheter **16**.

(81) As described above, fluid may be pumped continuously into imaging hood **12** to provide for clear viewing of the underlying tissue. Alternatively, fluid may be pumped temporarily or

sporadically only until a clear view of the tissue is available to be imaged and recorded, at which point the fluid flow may cease and the blood may be allowed to seep or flow back into imaging hood **12**. FIGS. **9A** to **9C** illustrate an example of capturing several images of the tissue at multiple regions. Deployment catheter **16** may be desirably positioned and imaging hood **12** deployed and brought into position against a region of tissue to be imaged, in this example the tissue surrounding a mitral valve MV within the left atrium of a patient's heart. The imaging hood **12** may be optionally anchored to the tissue, as described above, and then cleared by pumping the imaging fluid into the hood **12**. Once sufficiently clear, the tissue may be visualized and the image captured by control electronics **118**. The first captured image **140** may be stored and/or transmitted wirelessly **124** to a monitor **128** for viewing by the physician, as shown in FIG. **9A**.

(82) The deployment catheter **16** may be then repositioned to an adjacent portion of mitral valve MV, as shown in FIG. **9B**, where the process may be repeated to capture a second image **142** for viewing and/or recording. The deployment catheter **16** may again be repositioned to another region of tissue, as shown in FIG. **9C**, where a third image **144** may be captured for viewing and/or recording. This procedure may be repeated as many times as necessary for capturing a comprehensive image of the tissue surrounding mitral valve MV, or any other tissue region. When the deployment catheter **16** and imaging hood **12** is repositioned from tissue region to tissue region, the pump may be stopped during positioning and blood or surrounding fluid may be allowed to enter within imaging hood **12** until the tissue is to be imaged, where the imaging hood **12** may be cleared, as above.

(83) As mentioned above, when the imaging hood **12** is cleared by pumping the imaging fluid within for clearing the blood or other bodily fluid, the fluid may be pumped continuously to maintain the imaging fluid within the hood **12** at a positive pressure or it may be pumped under computer control for slowing or stopping the fluid flow into the hood **12** upon detection of various parameters or until a clear image of the underlying tissue is obtained. The control electronics **118** may also be programmed to coordinate the fluid flow into the imaging hood **12** with various physical parameters to maintain a clear image within imaging hood **12**.

(84) One example is shown in FIG. **10A** which shows a chart **150** illustrating how fluid pressure within die imaging hood **12** may be coordinated with the surrounding blood pressure. Chart **150** shows the cyclical blood pressure **156** alternating between diastolic pressure **152** and systolic pressure **154** over time T due to the beating motion of the patient heart. The fluid pressure of the imaging fluid, indicated by plot **160**, within imaging hood **12** may be automatically timed to correspond to the blood pressure changes **160** such that an increased pressure is maintained within imaging hood **12** which is consistently above the blood pressure **156** by a slight increase ΔP , as illustrated by the pressure difference at the peak systolic pressure **158**. This pressure difference, ΔP , may be maintained within imaging hood **12** over the pressure variance of the surrounding blood pressure to maintain a positive imaging fluid pressure within imaging hood **12** to maintain a clear view of the underlying tissue. One benefit of maintaining a constant ΔP is a constant flow and maintenance of a clear field

(85) FIG. **10B** shows a chart **162** illustrating another variation for maintaining a clear view of the underlying tissue where one or more sensors within the imaging hood **12**, as described in further detail below, may be configured to sense pressure changes within the imaging hood **12** and to correspondingly increase the imaging fluid pressure within imaging hood **12**. This may result in a time delay, ΔT , as illustrated by the shifted fluid pressure **160** relative to the cycling blood pressure **156**, although the time delay ΔT may be negligible in maintaining the clear image of the underlying tissue. Predictive software algorithms can also be used to substantially eliminate this time delay by predicting when the next pressure wave peak will arrive and by increasing the pressure ahead of die pressure wave's arrival by an amount of time equal to the aforementioned time delay to essentially cancel the time delay out.

(86) The variations in fluid pressure within imaging hood **12** may be accomplished in part due to

the nature of imaging hood **12**. An inflatable balloon, which is conventionally utilized for imaging tissue, may be affected by the surrounding blood pressure changes. On the other hand, an imaging hood **12** retains a constant volume therewithin and is structurally unaffected by the surrounding blood pressure changes, thus allowing for pressure increases therewithin. The material that hood **12** is made from may also contribute to the manner in which the pressure is modulated within this hood **12**. A stiffer hood material, such as high durometer polyurethane or Nylon, may facilitate the maintaining of an open hood when deployed. On the other hand, a relatively lower durometer or softer material, such as a low durometer PVC or polyurethane, may collapse from the surrounding fluid pressure and may not adequately maintain a deployed or expanded hood.

(87) Turning now to the imaging hood, other variations of the tissue imaging assembly may be utilized, as shown in FIG. **11A**, which shows another variation comprising an additional imaging balloon **172** within an imaging hood **174**. In this variation, an expandable balloon **172** having a translucent skin may be positioned within imaging hood **174**. Balloon **172** may be made from any distensible biocompatible material having sufficient translucent properties which allow for visualization therethrough. Once the imaging hood **174** has been deployed against the tissue region of interest, balloon **172** may be filled with a fluid, such as saline, or less preferably a gas, until balloon **172** has been expanded until the blood has been sufficiently displaced. The balloon **172** may this be expanded proximal to or into contact against the tissue region to be viewed. The balloon **172** can also be filled with contrast media to allow it to be viewed on fluoroscopy to aid in its positioning. The imager, e.g., fiber optic, positioned within deployment catheter **170** may then be utilized to view the tissue region through the balloon **172** and any additional fluid which may be pumped into imaging hood **174** via one or more optional fluid ports **176**, which may be positioned proximally of balloon **172** along a portion of deployment catheter **170**. Alternatively, balloon **172** may define one or more holes over its surface which allow for seepage or passage of the fluid contained therein to escape and displace the blood from within imaging hood **174**.

(88) FIG. **11B** shows another alternative in which balloon **180** may be utilized alone. Balloon **180**, attached to deployment catheter **178**, may be filled with fluid, such as saline or contrast media, and is preferably allowed to come into direct contact with the tissue region to be imaged.

(89) FIG. **12A** shows another alternative in which deployment catheter **16** incorporates imaging hood **12**, as above, and includes an additional flexible membrane **182** within imaging hood **12**. Flexible membrane **182** may be attached at a distal end of catheter **16** and optionally at contact edge **22**. Imaging hood **12** may be utilized, as above, and membrane **182** may be deployed from catheter **16** in vivo or prior to placing catheter **16** within a patient to reduce the volume within imaging hood **12**. The volume may be reduced or minimized to reduce the amount of fluid dispensed for visualization or simply be to reduced depending upon the area of tissue to be visualized.

(90) FIGS. **12B** and **12C** show yet another alternative in which imaging hood **186** may be withdrawn proximally within deployment catheter **184** or deployed distally from catheter **186**, as shown, to vary the volume of imaging hood **186** and thus the volume of dispensed fluid. Imaging hood **186** may be seen in FIG. **12B** as being partially deployed from, e.g., a circumferentially defined lumen within catheter **184**, such as annular lumen **188**. The underlying tissue may be visualized with imaging hood **186** only partially deployed. Alternatively, imaging hood **186'** may be fully deployed, as shown in FIG. **12C**, by urging hood **186'** distally out from annular lumen **188**. In this expanded configuration, the area of tissue to be visualized may be increased as hood **186'** is expanded circumferentially.

(91) FIGS. **13A** and **13B** show perspective and cross-sectional side views, respectively, of yet another variation of imaging assembly which may utilize a fluid suction system for minimizing the amount of fluid injected into the patient's heart or other body lumen during tissue visualization. Deployment catheter **190** in this variation may define an inner tubular member **196** which may be integrated with deployment catheter **190** or independently translatable. Fluid delivery lumen **198**

defined through member **196** may be fluidly connected to imaging hood **192**, which may also define one or more open channels **194** over its contact lip region. Fluid pumped through fluid delivery lumen **198** may thus fill open area **202** to displace any blood or other fluids or objects therewithin. As the clear fluid is forced out of open area **202**, it may be sucked or drawn immediately through one or more channels **194** and back into deployment catheter **190**. Tubular member **196** may also define one or more additional working channels **200** for the passage of any tools or visualization devices.

(92) In deploying the imaging hood in the examples described herein, the imaging hood may take on any number of configurations when positioned or configured for a low-profile delivery within the delivery catheter, as shown in the examples of FIGS. **14A** to **14D**. These examples are intended to be illustrative and are not intended to be limiting in scope. FIG. **14A** shows one example in which imaging hood **212** may be compressed within catheter **210** by folding hood **212** along a plurality of pleats. Hood **212** may also comprise scaffolding or frame **214** made of a super-elastic or shape memory material or alloy, e.g., Nitinol, Elgiloy, shape memory polymers, electroactive polymers, or a spring stainless steel. The shape memory material may act to expand or deploy imaging hood **212** into its expanded configuration when urged in the direction of the arrow from the constraints of catheter **210**.

(93) FIG. **14B** shows another example in which imaging hood **216** may be expanded or deployed from catheter **210** from a folded and overlapping configuration. Frame or scaffolding **214** may also be utilized in this example. FIG. **14C** shows yet another example in which imaging hood **218** may be rolled, inverted, or everted upon itself for deployment. In yet another example, FIG. **14D** shows a configuration in which imaging hood **220** may be fabricated from an extremely compliant material which allows for hood **220** to be simply compressed into a low-profile shape. From this low-profile compressed shape, simply releasing hood **220** may allow for it to expand into its deployed configuration, especially if a scaffold or frame of a shape memory or superelastic material, e.g., Nitinol, is utilized in its construction.

(94) Another variation for expanding the imaging hood is shown in FIGS. **15A** and **15B** which illustrates an helically expanding frame or support **230**. In its constrained low-profile configuration, shown in FIG. **15A**, helical frame **230** may be integrated with the imaging hood **12** membrane. When free to expand, as shown in FIG. **15B**, helical frame **230** may expand into a conical or tapered shape. Helical frame **230** may alternatively be made out of heat-activated Nitinol to allow it to expand upon application of a current.

(95) FIGS. **16A** and **16B** show yet another variation in which imaging hood **12** may comprise one or more hood support members **232** integrated with the hood membrane. These longitudinally attached support members **232** may be pivotably attached at their proximal ends to deployment catheter **16**. One or more pullwires **234** may be routed through the length of deployment catheter **16** and extend through one or more openings **238** defined in deployment catheter **16** proximally to imaging hood **12** into attachment with a corresponding support member **232** at a pullwire attachment point **236**. The support members **232** may be fabricated from a plastic or metal, such as stainless steel. Alternatively, the support members **232** may be made from a superelastic or shape memory alloy, such as Nitinol, which may self-expand into its deployed configuration without the use or need of pullwires. A heat-activated Nitinol may also be used which expands upon the application of thermal energy or electrical energy. In another alternative, support members **232** may also be constructed as inflatable lumens utilizing, e.g., PET balloons. From its low-profile delivery configuration shown in FIG. **16A**, the one or more pullwires **234** may be tensioned from their proximal ends outside the patient body to pull a corresponding support member **232** into a deployed configuration, as shown in FIG. **16B**, to expand imaging hood **12**. To reconfigure imaging hood **12** back into its low profile, deployment catheter **16** may be pulled proximally into a constraining catheter or the pullwires **234** may be simply pushed distally to collapse imaging hood **12**.

(96) FIGS. 17A and 17B show yet another variation of imaging hood 240 having at least two or more longitudinally positioned support members 242 supporting the imaging hood membrane. The support members 242 each have cross-support members 244 which extend diagonally between and are pivotably attached to the support members 242. Each of the cross-support members 244 may be pivotably attached to one another where they intersect between the support members 242. A jack or screw member 246 may be coupled to each cross-support member 244 at this intersection point and a torquing member, such as a torqueable wire 248, may be coupled to each jack or screw member 246 and extend proximally through deployment catheter 16 to outside the patient body. From outside the patient body, the torqueable wires 248 may be torqued to turn the jack or screw member 246 which in turn urges the cross-support members 244 to angle relative to one another and thereby urge the support members 242 away from one another. Thus, the imaging hood 240 may be transitioned from its low-profile, shown in FIG. 17A, to its expanded profile, shown in FIG. 17B, and back into its low-profile by torquing wires 248.

(97) FIGS. 18A and 18B show yet another variation on the imaging hood and its deployment. As shown, a distal portion of deployment catheter 16 may have several pivoting members 250, e.g., two to four sections, which form a tubular shape in its low profile configuration, as shown in FIG. 18A. When pivoted radially about deployment catheter 16, pivoting members 250 may open into a deployed configuration having distensible or expanding membranes 252 extending over the gaps in-between the pivoting members 250, as shown in FIG. 18B. The distensible membrane 252 may be attached to the pivoting members 250 through various methods, e.g., adhesives, such that when the pivoting members 250 are fully extended into a conical shape, the pivoting members 250 and membrane 252 form a conical shape for use as an imaging hood. The distensible membrane 252 may be made out of a porous material such as a mesh or PTFE or out of a translucent or transparent polymer such as polyurethane, PVC, Nylon, etc.

(98) FIGS. 19A and 19B show yet another variation where the distal portion of deployment catheter 16 may be fabricated from a flexible metallic or polymeric material to form a radially expanding hood 254. A plurality of slots 256 may be formed in a uniform pattern over the distal portion of deployment catheter 16, as shown in FIG. 19A. The slots 256 may be formed in a pattern such that when the distal portion is urged radially open, utilizing any of the methods described above, a radially expanded and conically-shaped hood 254 may be formed by each of the slots 256 expanding into an opening, as shown in FIG. 19B. A distensible membrane 258 may overlie the exterior surface or the interior surface of the hood 254 to form a fluid-impermeable hood 254 such that the hood 254 may be utilized as an imaging hood. Alternatively, the distensible membrane 258 may alternatively be formed in each opening 258 to form the fluid-impermeable hood 254. Once the imaging procedure has been completed, hood 254 may be retracted into its low-profile configuration.

(99) Yet another configuration for the imaging hood may be seen in FIGS. 20A and 20B where the imaging hood may be formed from a plurality of overlapping hood members 260 which overlie one another in an overlapping pattern. When expanded, each of the hood members 260 may extend radially outward relative to deployment catheter 16 to form a conically-shaped imaging hood, as shown in FIG. 20B. Adjacent hood members 260 may overlap one another along an overlapping interface 262 to form a fluid-retaining surface within the imaging hood. Moreover, the hood members 260 may be made from any number of biocompatible materials, e.g., Nitinol, stainless steel, polymers, etc., which are sufficiently strong to optionally retract surrounding tissue from the tissue region of interest.

(100) Although it is generally desirable to have an imaging hood contact against a tissue surface in a normal orientation, the imaging hood may be alternatively configured to contact the tissue surface at an acute angle. An imaging hood configured for such contact against tissue may also be especially suitable for contact against tissue surfaces having an unpredictable or uneven anatomical geography. For instance, as shown in the variation of FIG. 21A, deployment catheter 270 may have

an imaging hood **272** that is configured to be especially compliant. In this variation, imaging hood **272** may be comprised of one or more sections **274** that are configured to fold or collapse, e.g. by utilizing a pleated surface. Thus, as shown in FIG. **21B**, when imaging hood **272** is contacted against uneven tissue surface **T**, sections **274** are able to conform closely against the tissue. These sections **274** may be individually collapsible by utilizing an accordion style construction to allow conformation, e.g. to the trabeculae in the heart or the uneven anatomy that may be found inside the various body lumens.

(101) In yet another alternative, FIG. **22A** shows another variation in which an imaging hood **282** is attached to deployment catheter **280**. The contact lip or edge **284** may comprise one or more electrical contacts **286** positioned circumferentially around contact edge **284**. The electrical contacts **286** may be configured to contact the tissue and indicate affirmatively whether tissue contact was achieved, e.g., by measuring the differential impedance between blood and tissue. Alternatively, a processor, e.g., processor **98**, in electrical communication with contacts **286** may be configured to determine what type of tissue is in contact with electrical contacts **286**. In yet another alternative, the processor **98** may be configured to measure any electrical activity that may be occurring in the underlying tissue. e.g., accessory pathways, for the purposes of electrically mapping the cardiac tissue and subsequently treating, as described below, any arrhythmias which may be detected.

(102) Another variation for ensuring contact between imaging hood **282** and the underlying tissue may be seen in FIG. **22B**. This variation may have an inflatable contact edge **288** around the circumference of imaging hood **282**. The inflatable contact edge **288** may be inflated with a fluid or gas through inflation lumen **289** when the imaging hood **282** is to be placed against a tissue surface having an uneven or varied anatomy. The inflated circumferential surface **288** may provide for continuous contact over the hood edge by conforming against the tissue surface and facilitating imaging fluid retention within hood **282**.

(103) Aside from the imaging hood, various instrumentation may be utilized with the imaging and manipulation system. For instance, after the field within imaging hood **12** has been cleared of the opaque blood and the underlying tissue is visualized through the clear fluid, blood may seep back into the imaging hood **12** and obstruct the view. One method for automatically maintaining a clear imaging field may utilize a transducer, e.g., an ultrasonic transducer **290**, positioned at the distal end of deployment catheter within the imaging hood **12**, as shown in FIG. **23**. The transducer **290** may send an energy pulse **292** into the imaging hood **12** and wait to detect back-scattered energy **294** reflected from debris or blood within the imaging hood **12**. If back-scattered energy is detected, the pump may be actuated automatically to dispense more fluid into the imaging hood until the debris or blood is no longer detected.

(104) Alternatively, one or more sensors **300** may be positioned on the imaging hood **12** itself, as shown in FIG. **24A**, to detect a number of different parameters. For example, sensors **300** may be configured to detect for the presence of oxygen in the surrounding blood, blood and/or imaging fluid pressure, color of the fluid within the imaging hood, etc. Fluid color may be particularly useful in detecting the presence of blood within the imaging hood **12** by utilizing a reflective type sensor to detect back reflection from blood. Any reflected light from blood which may be present within imaging hood **12** may be optically or electrically transmitted through deployment catheter **16** and to a red colored filter within control electronics **118**. Any red color which may be detected may indicate the presence of blood and trigger a signal to the physician or automatically actuate the pump to dispense more fluid into the imaging hood **12** to clear the blood.

(105) Alternative methods for detecting the presence of blood within the hood **12** may include detecting transmitted light through the imaging fluid within imaging hood **12**. If a source of white light, e.g., utilizing LEDs or optical fibers, is illuminated inside imaging hood **12**, the presence of blood may cause the color red to be filtered through this fluid. The degree or intensity of the red color detected may correspond to the amount of blood present within imaging hood **12**. A red color

sensor can simply comprise, in one variation, a phototransistor with a red transmitting filter over it which can establish how much red light is detected, which in turn can indicate the presence of blood within imaging hood **12**. Once blood is detected, the system may pump more clearing fluid through and enable closed loop feedback control of the clearing fluid pressure and flow level. (106) Any number of sensors may be positioned along the exterior **302** of imaging hood **12** or within the interior **304** of imaging hood **12** to detect parameters not only exteriorly to imaging hood **12** but also within imaging hood **12**. Such a configuration, as shown in FIG. **24B**, may be particularly useful for automatically maintaining a clear imaging field based upon physical parameters such as blood pressure, as described above for FIGS. **10A** and **10B**.

(107) Aside from sensors, one or more light emitting diodes (LEDs) may be utilized to provide lighting within the imaging hood **12**. Although illumination may be provided by optical fibers routed through deployment catheter **16**, the use of LEDs over the imaging hood **12** may eliminate the need for additional optical fibers for providing illumination. The electrical wires connected to the one or more LEDs may be routed through or over the hood **12** and along an exterior surface or extruded within deployment catheter **16**. One or more LEDs may be positioned in a circumferential pattern **306** around imaging hood **12**, as shown in FIG. **25A**, or in a linear longitudinal pattern **308** along imaging hood **12**, as shown in FIG. **25B**. Other patterns, such as a helical or spiral pattern, may also be utilized. Alternatively, LEDs may be positioned along a support member forming part of imaging hood **12**.

(108) In another alternative for illumination within imaging hood **12**, a separate illumination tool **310** may be utilized, as shown in FIG. **26A**. An example of such a tool may comprise a flexible intravascular delivery member **312** having a carrier member **314** pivotably connected **316** to a distal end of delivery member **312**. One or more LEDs **318** may be mounted along carrier member **314**. In use, delivery member **312** may be advanced through deployment catheter **16** until carrier member **314** is positioned within imaging hood **12**. Once within imaging hood **12**, carrier member **314** may be pivoted in any number of directions to facilitate or optimize the illumination within the imaging hood **12**, as shown in FIG. **26B**.

(109) In utilizing LEDs for illumination, whether positioned along imaging hood **12** or along a separate instrument, the LEDs may comprise a single LED color, e.g., white light. Alternatively, LEDs of other colors, e.g., red, blue, yellow, etc., may be utilized exclusively or in combination with white LEDs to provide for varied illumination of the tissue or fluids being imaged. Alternatively, sources of infrared or ultraviolet light may be employed to enable imaging beneath the tissue surface or cause fluorescence of tissue for use in system guidance, diagnosis, or therapy.

(110) Aside from providing a visualization platform, the imaging assembly may also be utilized to provide a therapeutic platform for treating tissue being visualized. As shown in FIG. **27**, deployment catheter **320** may have imaging hood **322**, as described above, and fluid delivery lumen **324** and imaging lumen **326**. In this variation, a therapeutic tool such as needle **328** may be delivered through fluid delivery lumen **324** or in another working lumen and advanced through open area **332** for treating the tissue which is visualized. In this instance, needle **328** may define one or several ports **330** for delivering drugs therethrough. Thus, once the appropriate region of tissue has been imaged and located, needle **328** may be advanced and pierced into the underlying tissue where a therapeutic agent may be delivered through ports **330**. Alternatively, needle **328** may be in electrical communication with a power source **334**, e.g., radio-frequency, microwave, etc., for ablating the underlying tissue area of interest.

(111) FIG. **28** shows another alternative in which deployment catheter **340** may have imaging hood **342** attached thereto, as above, but with a therapeutic tool **344** in the configuration of a helical tissue piercing device **344**. Also shown and described above in FIGS. **7A** and **7B** for use in stabilizing the imaging hood relative to the underlying tissue, the helical tissue piercing device **344** may also be utilized to manipulate the tissue for a variety of therapeutic procedures. The helical portion **346** may also define one or several ports for delivery of therapeutic agents therethrough.

(112) In yet another alternative, FIG. 29 shows a deployment catheter 350 having an expandable imaging balloon 352 filled with, e.g., saline 356. A therapeutic tool 344, as above, may be translatable relative to balloon 352. To prevent the piercing portion 346 of the tool from tearing balloon 352, a stop 354 may be formed on balloon 352 to prevent the proximal passage of portion 346 past stop 354.

(113) Alternative configurations for tools which may be delivered through deployment catheter 16 for use in tissue manipulation within imaging hood 12 are shown in FIGS. 30A and 30B. FIG. 30A shows one variation of an angled instrument 360, such as a tissue grasper, which may be configured to have an elongate shaft for intravascular delivery through deployment catheter 16 with a distal end which may be angled relative to its elongate shaft upon deployment into imaging hood 12. The elongate shaft may be configured to angle itself automatically, e.g., by the elongate shaft being made at least partially from a shape memory alloy, or upon actuation. e.g., by tensioning a pullwire. FIG. 30B shows another configuration for an instrument 362 being configured to reconfigure its distal portion into an off-axis configuration within imaging hood 12. In either case, the instruments 360, 362 may be reconfigured into a low-profile shape upon withdrawing them proximally back into deployment catheter 16.

(114) Other instruments or tools which may be utilized with the imaging system is shown in the side and end views of FIGS. 31A to 31C. FIG. 31A shows a probe 370 having a distal end effector 372, which may be reconfigured from a low-profile shape to a curved profile. The end effector 372 may be configured as an ablation probe utilizing radio-frequency energy, microwave energy, ultrasound energy, laser energy or even cryo-ablation. Alternatively, the end effector 372 may have several electrodes upon it for detecting or mapping electrical signals transmitted through the underlying tissue.

(115) In the case of an end effector 372 utilized for ablation of the underlying tissue, an additional temperature sensor such as a thermocouple or thermistor 374 positioned upon an elongate member 376 may be advanced into the imaging hood 12 adjacent to the distal end effector 372 for contacting and monitoring a temperature of the ablated tissue. FIG. 31B shows an example in the end view of one configuration for the distal end effector 372 which may be simply angled into a perpendicular configuration for contacting the tissue. FIG. 31C shows another example where the end effector may be reconfigured into a curved end effector 378 for increased tissue contact.

(116) FIGS. 32A and 32B show another variation of an ablation tool utilized with an imaging hood 12 having an enclosed bottom portion. In this variation, an ablation probe, such as a cryo-ablation probe 380 having a distal end effector 382, may be positioned through the imaging hood 12 such that the end effector 382 is placed distally of a transparent membrane or enclosure 384, as shown in the end view of FIG. 32B. The shaft of probe 380 may pass through an opening 386 defined through the membrane 384. In use, the clear fluid may be pumped into imaging hood 12, as described above, and the distal end effector 382 may be placed against a tissue region to be ablated with the imaging hood 12 and the membrane 384 positioned atop or adjacent to the ablated tissue. In the case of cryo-ablation, the imaging fluid may be warmed prior to dispensing into the imaging hood 12 such that the tissue contacted by the membrane 384 may be warmed during the cryo-ablation procedure. In the case of thermal ablation, e.g., utilizing radio-frequency energy, the fluid dispensed into the imaging hood 12 may be cooled such that the tissue contacted by the membrane 384 and adjacent to the ablation probe during the ablation procedure is likewise cooled.

(117) In either example described above, the imaging fluid may be varied in its temperature to facilitate various procedures to be performed upon the tissue. In other cases, the imaging fluid itself may be altered to facilitate various procedures. For instance as shown in FIG. 33A, a deployment catheter 16 and imaging hood 12 may be advanced within a hollow body organ, such as a bladder filled with urine 394, towards a lesion or tumor 392 on the bladder wall. The imaging hood 12 may be placed entirely over the lesion 392, or over a portion of the lesion. Once secured against the tissue wall 390, a cryo-fluid, i.e., a fluid which has been cooled to below freezing temperatures of,

e.g., water or blood, may be pumped into the imaging hood **12** to cryo-ablate the lesion **390**, as shown in FIG. **33B** while avoiding the creation of ice on the instrument or surface of tissue.

(118) As the cryo-fluid leaks out of the imaging hood **12** and into the organ, the fluid may be warmed naturally by the patient body and ultimately removed. The cryo-fluid may be a colorless and translucent fluid which enables visualization therethrough of the underlying tissue. An example of such a fluid is Fluorinert™ (3M, St. Paul, Minn.), which is a colorless and odorless perfluorinated liquid. The use of a liquid such as Fluorinert™ enables the cryo-ablation procedure without the formation of ice within or outside of the imaging hood **12**. Alternatively, rather than utilizing cryo-ablation, hyperthermic treatments may also be effected by heating the Fluorinert™ liquid to elevated temperatures for ablating the lesion **392** within the imaging hood **12**. Moreover, Fluorinert™ may be utilized in various other parts of the body, such as within the heart.

(119) FIG. **34A** shows another variation of an instrument which may be utilized with the imaging system. In this variation, a laser ring generator **400** may be passed through the deployment catheter **16** and partially into imaging hood **12**. A laser ring generator **400** is typically used to create a circular ring of laser energy **402** for generating a conduction block around the pulmonary veins typically in the treatment of atrial fibrillation. The circular ring of laser energy **402** may be generated such that a diameter of the ring **402** is contained within a diameter of the imaging hood **12** to allow for tissue ablation directly upon tissue being imaged. Signals which cause atrial fibrillation typically come from the entry area of the pulmonary veins into the left atrium and treatments may sometimes include delivering ablation energy to the ostia of the pulmonary veins within the atrium. The ablated areas of the tissue may produce a circular scar which blocks the impulses for atrial fibrillation.

(120) When using the laser energy to ablate the tissue of the heart, it may be generally desirable to maintain the integrity and health of the tissue overlying the surface while ablating the underlying tissue. This may be accomplished, for example, by cooling the imaging fluid to a temperature below the body temperature of the patient but which is above the freezing point of blood (e.g., 2.degree. C. to 35.degree. C.). The cooled imaging fluid may thus maintain the surface tissue at the cooled fluid temperature while the deeper underlying tissue remains at the patient body temperature. When the laser energy (or other types of energy such as radio frequency energy, microwave energy, ultrasound energy, etc.) irradiates the tissue, both the cooled tissue surface as well as the deeper underlying tissue will rise in temperature uniformly. The deeper underlying tissue, which was maintained at the body temperature, will increase to temperatures which are sufficiently high to destroy the underlying tissue. Meanwhile, the temperature of the cooled surface tissue will also rise but only to temperatures that are near body temperature or slightly above.

(121) Accordingly, as shown in FIG. **34B**, one example for treatment may include passing deployment catheter **16** across the atrial septum AS and into the left atrium LA of the patient's heart H. Other methods of accessing the left atrium LA may also be utilized. The imaging hood **12** and laser ring generator **400** may be positioned adjacent to or over one or more of the ostium OT of the pulmonary veins PV and the laser generator **400** may ablate the tissue around the ostium OT with the circular ring of laser energy **402** to create a conduction block. Once one or more of the tissue around the ostium OT have been ablated, the imaging hood **12** may be reconfigured into a low profile for removal from the patient heart H.

(122) One of the difficulties in treating tissue in or around the ostium OT is the dynamic fluid flow of blood through the ostium OT. The dynamic forces make cannulation or entry of the ostium OT difficult. Thus, another variation on instruments or tools utilizable with the imaging system is an extendible cannula **410** having a cannula lumen **412** defined therethrough, as shown in FIG. **35A**. The extendible cannula **410** may generally comprise an elongate tubular member which may be positioned within the deployment catheter **16** during delivery and then projected distally through the imaging hood **12** and optionally beyond, as shown in FIG. **35B**.

(123) In use, once the imaging hood **12** has been desirably positioned relative to the tissue, e.g., as

shown in FIG. 35C outside the ostium OT of a pulmonary vein PV, the extendible cannula **410** may be projected distally from the deployment catheter **16** while optionally imaging the tissue through the imaging hood **12**, as described above. The extendible cannula **410** may be projected distally until its distal end is extended at least partially into the ostium OT. Once in the ostium OT, an instrument or energy ablation device may be extended through and out of the cannula lumen **412** for treatment within the ostium OT. Upon completion of the procedure, the cannula **410** may be withdrawn proximally and removed from the patient body. The extendible cannula **410** may also include an inflatable occlusion balloon at or near its distal end to block the blood flow out of the PV to maintain a clear view of the tissue region. Alternatively, the extendible cannula **410** may define a lumen therethrough beyond the occlusion balloon to bypass at least a portion of the blood that normally exits the pulmonary vein PV by directing the blood through the cannula **410** to exit proximal of the imaging hood.

(124) Yet another variation for tool or instrument use may be seen in the side and end views of FIGS. 36A and 36B. In this variation, imaging hood **12** may have one or more tubular support members **420** integrated with the hood **12**. Each of the tubular support members **420** may define an access lumen **422** through which one or more instruments or tools may be delivered for treatment upon the underlying tissue. One particular example is shown and described above for FIG. 7C.

(125) Various methods and instruments may be utilized for using or facilitating the use of the system. For instance, one method may include facilitating the initial delivery and placement of a device into the patient's heart. In initially guiding the imaging assembly within the heart chamber to, e.g., the mitral valve MV, a separate guiding probe **430** may be utilized, as shown in FIGS. 37A and 37B. Guiding probe **430** may, for example, comprise an optical fiber through which a light source **434** may be used to illuminate a distal tip portion **432**. The tip portion **432** may be advanced into the heart through, e.g., the coronary sinus CS, until the tip is positioned adjacent to the mitral valve MV. The tip **432** may be illuminated, as shown in FIG. 37A, and imaging assembly **10** may then be guided towards the illuminated tip **432**, which is visible from within the atrial chamber, towards mitral valve MV.

(126) Aside from the devices and methods described above, the imaging system may be utilized to facilitate various other procedures. Turning now to FIGS. 38A and 38B, the imaging hood of the device in particular may be utilized. In this example, a collapsible membrane or disk-shaped member **440** may be temporarily secured around the contact edge or lip of imaging hood **12**. During intravascular delivery, the imaging hood **12** and the attached member **440** may both be in a collapsed configuration to maintain a low profile for delivery. Upon deployment, both the imaging hood **12** and the member **440** may extend into their expanded configurations.

(127) The disk-shaped member **440** may be comprised of a variety of materials depending upon the application. For instance, member **440** may be fabricated from a porous polymeric material infused with a drug eluting medicament **442** for implantation against a tissue surface for slow infusion of the medicament into the underlying tissue. Alternatively, the member **440** may be fabricated from a non-porous material, e.g., metal or polymer, for implantation and closure of a wound or over a cavity to prevent fluid leakage. In yet another alternative, the member **440** may be made from a distensible material which is secured to imaging hood **12** in an expanded condition. Once implanted or secured on a tissue surface or wound, the expanded member **440** may be released from imaging hood **12**. Upon release, the expanded member **440** may shrink to a smaller size while approximating the attached underlying tissue, e.g., to close a wound or opening.

(128) One method for securing the disk-shaped member **440** to a tissue surface may include a plurality of tissue anchors **444**, e.g., barbs, hooks, projections, etc., which are attached to a surface of the member **440**. Other methods of attachments may include adhesives, suturing, etc. In use, as shown in FIGS. 39A to 39C, the imaging hood **12** may be deployed in its expanded configuration with member **440** attached thereto with the plurality of tissue anchors **444** projecting distally. The tissue anchors **444** may be urged into a tissue region to be treated **446**, as seen in FIG. 39A, until

the anchors **444** are secured in the tissue and member **440** is positioned directly against the tissue, as shown in FIG. **39B**. A pullwire may be actuated to release the member **440** from the imaging hood **12** and deployment catheter **16** may be withdrawn proximally to leave member **440** secured against the tissue **446**.

(129) Another variation for tissue manipulation and treatment may be seen in the variation of FIG. **40A**, which illustrates an imaging hood **12** having a deployable anchor assembly **450** attached to the tissue contact edge **22**. FIG. **40B** illustrates the anchor assembly **450** detached from the imaging hood **12** for clarity. The anchor assembly **450** may be seen as having a plurality of discrete tissue anchors **456**, e.g., barbs, hooks, projections, etc., each having a suture retaining end, e.g., an eyelet or opening **458** in a proximal end of the anchors **456**. A suture member or wire **452** may be slidably connected to each anchor **456** through the openings **458** and through a cinching element **454**, which may be configured to slide uni-directionally over the suture or wire **452** to approximate each of the anchors **456** towards one another. Each of the anchors **456** may be temporarily attached to the imaging hood **12** through a variety of methods. For instance, a pullwire or retaining wire may hold each of the anchors within a receiving ring around the circumference of the imaging hood **12**. When the anchors **456** are released, the pullwire or retaining wire may be tensioned from its proximal end outside the patient body to thereby free the anchors **456** from the imaging hood **12**.

(130) One example for use of the anchor assembly **450** is shown in FIGS. **41A** to **41D** for closure of an opening or wound **460**, e.g., patent foramen ovale (PFO). The deployment catheter **16** and imaging hood **12** may be delivered intravascularly into, e.g., a patient heart. As the imaging hood **12** is deployed into its expanded configuration, the imaging hood **12** may be positioned adjacent to the opening or wound **460**, as shown in FIG. **41A**. With the anchor assembly **450** positioned upon the expanded imaging hood **12**, deployment catheter **16** may be directed to urge the contact edge of imaging hood **12** and anchor assembly **450** into the region surrounding the tissue opening **460**, as shown in FIG. **41B**. Once the anchor assembly **450** has been secured within the surrounding tissue, the anchors may be released from imaging hood **12** leaving the anchor assembly **450** and suture member **452** trailing from the anchors, as shown in FIG. **41C**. The suture or wire member **452** may be tightened by pulling it proximally from outside the patient body to approximate the anchors of anchor assembly **450** towards one another in a purse-string manner to close the tissue opening **462**, as shown in FIG. **41D**. The cinching element **454** may also be pushed distally over the suture or wire member **452** to prevent the approximated anchor assembly **450** from loosening or widening.

(131) Another example for an alternative use is shown in FIG. **42**, where the deployment catheter **16** and deployed imaging hood **12** may be positioned within a patient body for drawing blood **472** into deployment catheter **16**. The drawn blood **472** may be pumped through a dialysis unit **470** located externally of the patient body for filtering the drawn blood **472** and the filtered blood may be reintroduced back into the patient.

(132) Yet another variation is shown in FIGS. **43A** and **43B**, which show a variation of the deployment catheter **480** having a first deployable hood **482** and a second deployable hood **484** positioned distal to the first hood **482**. The deployment catheter **480** may also have a side-viewing imaging element **486** positioned between the first and second hoods **482**, **484** along the length of the deployment catheter **480**. In use, such a device may be introduced through a lumen **488** of a vessel VS, where one or both hoods **482**, **484** may be expanded to gently contact the surrounding walls of vessel VS. Once hoods **482**, **484** have been expanded, the clear imaging fluid may be pumped in the space defined between the hoods **482**, **484** to displace any blood and to create an imaging space **490**, as shown in FIG. **43B**. With the clear fluid in-between hoods **482**, **484**, the imaging element **486** may be used to view the surrounding tissue surface contained between hoods **482**, **484**. Other instruments or tools may be passed through deployment catheter **480** and through one or more openings defined along the catheter **480** for additionally performing therapeutic procedures upon the vessel wall.

(133) Another variation of a deployment catheter **500** which may be used for imaging tissue to the

side of the instrument may be seen in FIGS. 44A to 45C. FIGS. 44A and 44B show side and end views of deployment catheter 500 having a side-imaging balloon 502 in an un-inflated low-profile configuration. A side-imaging element 504 may be positioned within a distal portion of the catheter 500 where the balloon 502 is disposed. When balloon 502 is inflated, it may expand radially to contact the surrounding tissue, but where the imaging element 504 is located, a visualization field 506 may be created by the balloon 502, as shown in the side, top, and end views of FIGS. 45A to 45C, respectively. The visualization field 506 may simply be a cavity or channel which is defined within the inflated balloon 502 such that the visualization element 504 is provided an image of the area within field 506 which is clear and unobstructed by balloon 502.

(134) In use, deployment catheter 500 may be advanced intravascularly through vessel lumen 488 towards a lesion or tumor 508 to be visualized and/or treated. Upon reaching the lesion 508, deployment catheter 500 may be positioned adjacently to the lesion 508 and balloon 502 may be inflated such that the lesion 508 is contained within the visualization field 506. Once balloon 502 is fully inflated and in contact against the vessel wall, clear fluid may be pumped into visualization field 506 through deployment catheter 500 to displace any blood or opaque fluids from the field 506, as shown in the side and end views of FIGS. 46A and 46B, respectively. The lesion 508 may then be visually inspected and treated by passing any number of instruments through deployment catheter 500 and into field 506.

(135) The applications of the disclosed invention discussed above are not limited to certain treatments or regions of the body, but may include any number of other treatments and areas of the body. Modification of the above-described methods and devices for carrying out the invention, and variations of aspects of the invention that are obvious to those of skill in the arts are intended to be within the scope of this disclosure. Moreover, various combinations of aspects between examples are also contemplated and are considered to be within the scope of this disclosure as well.

Claims

1. A tissue imaging assembly comprising: a first steerable catheter comprising a plurality of pull wires configured to control articulation of the first steerable catheter and a lumen, wherein the lumen opens through a distal face at a distal end of the first steerable catheter; and a second steerable catheter deployable within the lumen of the first steerable catheter, the second steerable catheter comprising: a lumen for delivering an elongate instrument out a distal region of the second steerable catheter; and an imager attached to the distal region of the second steerable catheter.
2. The tissue imaging assembly of claim 1, wherein the elongate instrument comprises a guidewire.
3. The tissue imaging assembly of claim 1, wherein the elongate instrument comprises a needle.
4. The tissue imaging assembly of claim 1, wherein the elongate instrument comprises an electrode.
5. The tissue imaging assembly of claim 1, wherein the imager comprises an optical imaging fiber.
6. The tissue imaging assembly of claim 1, wherein the second steerable catheter further comprises an expandable hood.
7. A method comprising: translating a second steerable catheter with respect to a first steerable catheter, the second steerable catheter supporting an imager attached to a distal region of the second steerable catheter and comprising a lumen for delivering an elongate instrument out the distal region of the second steerable catheter, the first steerable catheter comprising a plurality of pull wires configured to control articulation of the first steerable catheter and a lumen receiving the second steerable catheter, the lumen of the first steerable catheter opening through a distal face at a distal end of the first steerable catheter; inserting the elongate instrument through the lumen of the second steerable catheter; and imaging a tissue region with the imager while the elongate instrument projects distally from the lumen of the second steerable catheter.
8. The method of claim 7, further comprising, expanding an imaging hood that is at a distal end of the second steerable catheter, the imaging hood being independently translatable from the first

steerable catheter.

9. The method of claim 7, wherein the lumen of the second steerable catheter comprises a guidewire lumen and the elongate instrument comprises a guidewire.
